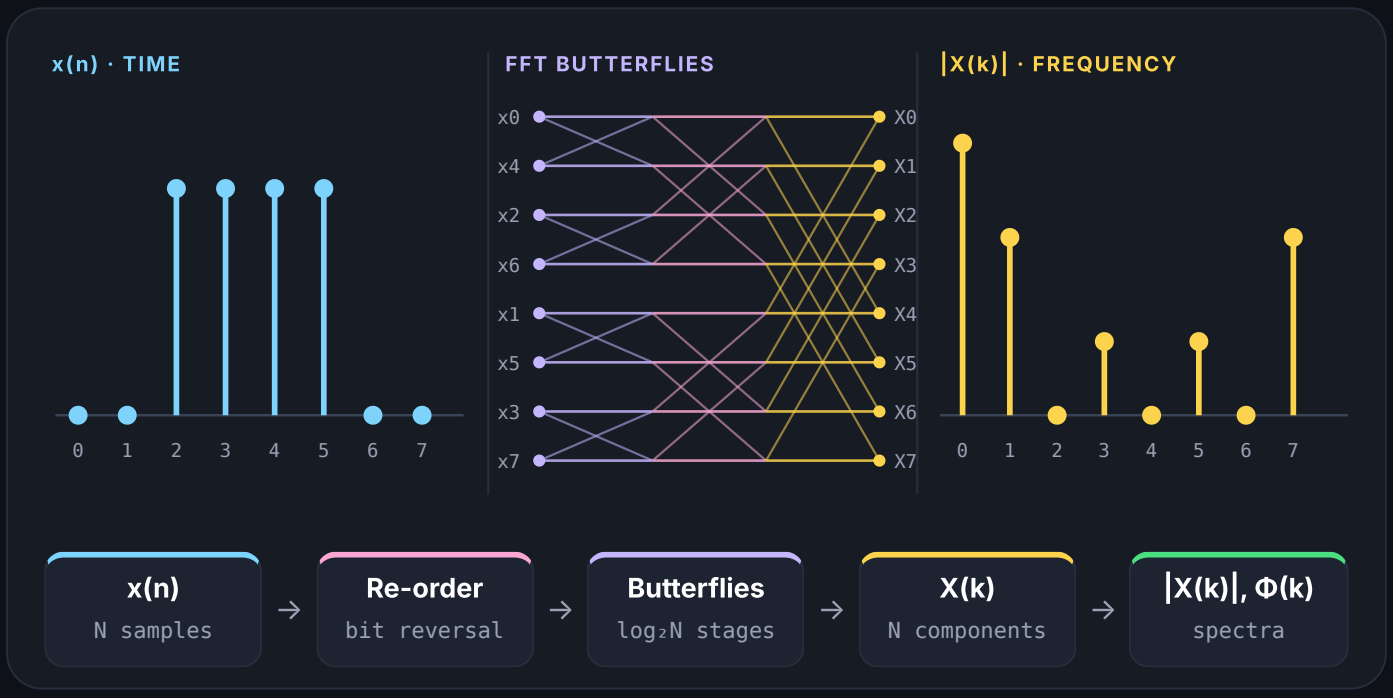
02

CHAPTER

The Discrete & Fast Fourier Transform

The Complete Chapter 2 Pack

DFT, IDFT, FFT and IFFT from the lectures, previous exams, Sheet 2 and the assignment — solved the doctor's way



13 theory questions

15 solved problems

6 Sheet 2 questions

19 previous exams & quizzes

PART A
Theory
Lectures 5 · 6

PART B
Problems
Method · Examples · Exams

PART C
Sheet 2
Problems · Assignment

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Modern Academy for Engineering & Technology

DSP / ch02 / contents

Contents

Theory · Problems · Sheet 2 · Assignment · 33 questions

A Theory Lectures 5 and 6

01 **Lecture 5** — DFT, IDFT & Properties — 8 questions — p. 4

02 **Lecture 6** — FFT, Butterfly & IFFT — 5 questions — p. 11

B Problems Solving method, lecture examples and previous exams

03 **Solving Method** — The doctor's steps for every problem type — 4 questions — p. 17

04 **Lecture Examples** — Examples 2.1 to 2.4, solved step by step — 4 questions — p. 19

05 **Previous Exams** — Midterms, finals & Revision Sheet — 5 questions — p. 26

C Sheet 2 & Assignment Theory, problems and the assignment

06 **Sheet 2 Theory** — Question 2.1 — 1 questions — p. 52

07 **Sheet 2 Problems** — Questions 2.2 to 2.6 — 5 questions — p. 54

08 **Assignment** — DSP Lecture Assignment, Fall 27 — 1 questions — p. 65

How to study from this file

A: theory, most repeated first. B: start with the solving method, then examples and exams. C: all of Sheet 2 and the assignment.

DSP / ch02 / repeated

Most Repeated Theory Questions

Most repeated in previous exams — start with these

Times asked	Question
13	Q3. Prove that the DFT, X(k), of a sequence x(n) is periodic.
10	P9. x(n) = {1, 0, 1, 1} — DFT / FFT / IFFT / Parseval / spectra / saving
4	P10. x(n) = {1, 0, 1, 0} — DFT / FFT / IFFT / Parseval / spectra / saving
3	P12. x(n) = {1, 0, 0, 1} — DFT / FFT / IFFT / Parseval / spectra / saving
3	Q8. Check (✓) or (×) and correct the mistakes (DFT / FT statements).

SECTION 01

Lecture 5

DFT, IDFT & Properties

Questions — most frequent first

- ☐ Q3 Prove that the DFT, X(k), of a sequence x(n) is periodic. p. 4
- ☐ Q8 Check (✓) or (×) and correct the mistakes (DFT / FT statements). p. 5
- ☐ Q1 Define the Discrete Fourier Transform (DFT). p. 6
- ☐ Q6 State the mathematical properties of the DFT. p. 7
- ☐ Q2 Find the relation between the DFT and the Fourier transform. p. 7
- ☐ Q4 Show that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2... p. 8
- ☐ Q5 Define the Inverse Discrete Fourier Transform (IDFT). p. 9
- ☐ Q7 State Parseval's theorem. p. 9

Q3 ● Frequent 13×

Prove that the DFT, X(k), of a sequence x(n) is periodic.

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}$$

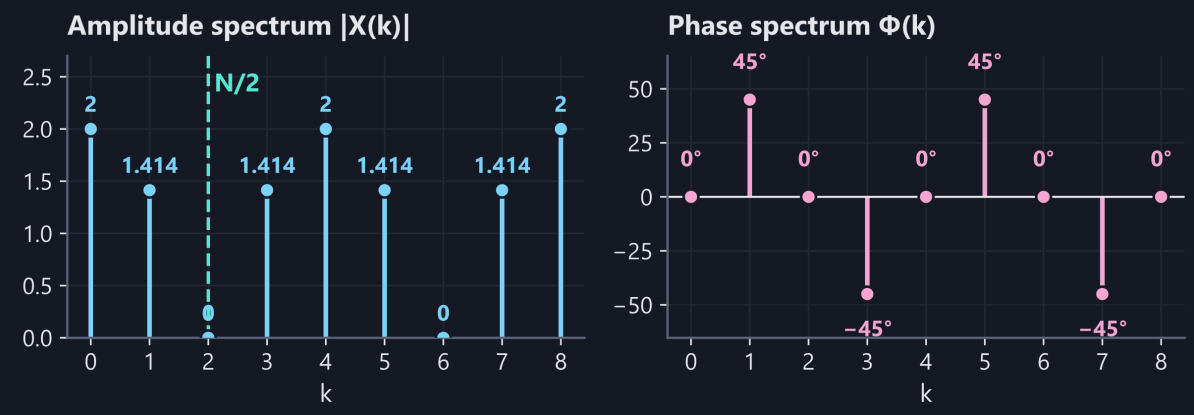
$$X(k+N) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi n}$$

Since n is integral, $e^{-j2\pi n} = 1$, so:

$$\mathbf{X(k+N) = X(k)}$$

The DFT is periodic with period N (the cyclical property of the DFT). E.g. for {1, 0, 0, 1}: X(0) = 2, therefore X(4) = 2 also.

|X(k)| for {1, 0, 0, 1}: X(k + 4) = X(k), symmetric about N/2



Key idea لأن $e^{(-j2\pi n)} = 1$ لأي n صحيح، لما نعوّض k + N بنرجع لنفس القيمة، يعني $X(k+N) = X(k)$ والـ DFT بيتكرر كل N.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 · p. 4

Came in

Final 7-6-2023 · Q1-b | Final 9-6-2026 · Q2-a | Final 17-8-2026 · Q2-a | Final 23-1-2023 · Q3-a | Final 12-1-2019 · Q2-b | Final 2-1-2017 · Q2-b | Final 23-12-2017 · Q2-b | Midterm 12-11-2025 · Q2-a | Midterm 28-11-2021 · Q2-a | Midterm 8-12-2020 · Q2-b | Midterm 1 Model-2 · Q2-a | Midterm 2 · Q2-b | Sheet 2 · 2.1

Q8 3×

Check (✓) or (×) and correct the mistakes (DFT / FT statements).

1. Fourier transform (FT) is calculated for non-periodic continuous-time signals. **(✓)**
2. Discrete Fourier transform (DFT) is obtained for the analog signals. **(×)**
   Correct: the DFT is obtained for the sampled (discrete-time) sequence x(nT).

Key idea الـ FT للإشارات الـ continuous اللي مش periodic، فدي صح. والـ DFT بيتحسب على الـ samples مش على الإشارة الأنالوج، فالتانية غلط.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 · p. 4

Came in Final 12-1-2019 · Q5 | Final 23-12-2017 · Q5-b | Quiz 1 Summer 26 · Q3

Q1 1×

Define the Discrete Fourier Transform (DFT).

Assume a waveform has been sampled at regular time intervals T to produce the sample sequence {x(nT)} = x(0), x(T), ..., x[(N − 1)T] of N sample values. The DFT of x(nT) is defined as the sequence of complex values {X(kΩ)} = X(0), X(Ω), ..., X[(N − 1)Ω] in the frequency domain, where Ω = 2π/NT is the first harmonic frequency:

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \quad (2.4)$$

$$X(k) = R(k) + jI(k) \quad (2.1) \qquad |X(k)| = [R^2(k) + I^2(k)]^{\frac{1}{2}} \quad (2.2) \qquad \Phi(k) = \tan^{-1}[I(k)/R(k)] \quad (2.3)$$

Key idea الـ DFT بياخد N samples في الوقت ويطلّع N مركبة complex في التردد، والمسافة بين كل مركبة والتانية Ω = 2π/NT.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 · p. 4

Came in Book exam Midterm 2015

Q6 1×

State the mathematical properties of the DFT.

1. **Symmetry:** Re[X(N − k)] = Re X(k) (2.8) and Im[X(N − k)] = −Im[X(k)] (2.9).
2. **Even functions:** if $x_e(n) = x_e(-n)$: $X_e(k) = \Sigma\, x_e(n) \cos(k\Omega nT)$ (2.10).
3. **Odd functions:** if $x_o(n) = -x_o(-n)$: $X_o(k) = -j\,\Sigma\, x_o(n) \sin(k\Omega nT)$ (2.11).
4. **Parseval's theorem:** $\Sigma\, x^2(n) = (1/N)\,\Sigma\, |X(k)|^2$ (2.12).
5. **Delta function:** $F_D[\delta(nT)] = 1$ (2.13).
6. Linear cross-correlation of two sequences may be computed using DFTs.
7. DFTs may be used to compute circular convolutions and, using augmenting zeros, linear convolutions.

Key idea احفظهم كقايمة: symmetry، و even و odd functions، و Parseval، و delta function، و linear cross-correlation، و circular convolution.

Key words: Symmetry | Even functions | Odd functions | Parseval's theorem

Source — from your PDFs

Lecture	**LEC 5** · Sec. 2.3 · p. 27
Came in	Final 18-1-2024 · Q2-a

Q2 Less frequent

Find the relation between the DFT and the Fourier transform.

Equation (2.4) is analogous to the Fourier transform $F(j\omega) = \int f(t)\, e^{-j\omega t}\, dt$ (F-1). Putting $x(nT) = f(t)$, $k\Omega = \omega$, $nT = t$, $dt = T$, and replacing the integral by a summation (for $f(t) = 0$ outside $0 \le t \le (N-1)T$):

$$\Sigma_{n=0}^{N-1}\, x(nT)\, e^{-jk\Omega nT}\, T = F(j\omega) \quad (2.5)$$

$$\mathbf{F(j\omega) = X(k)\ T} \quad \text{i.e.} \quad \mathbf{FT = DFT \times T} \quad (2.6)$$

The FT components are obtained by multiplying the DFT components by the sampling interval T. Similarly, **IFT = IDFT / T**.

Key idea لو قطّعنا تكامل الـ Fourier transform لـ samples بنوصل للـ DFT، فالعلاقة FT = DFT × T، والـ IFT = IDFT ÷ T.

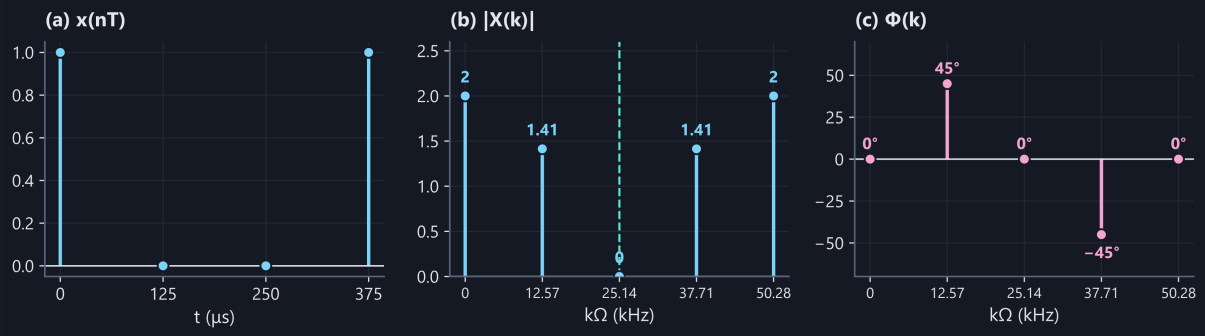
Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 · p. 4

Q4 Less frequent

Show that the amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2 (folding frequency).

- The amplitude spectrum of an N-point DFT is symmetrical about harmonic N/2, and the phase angles are an odd function centred round this component.
- If $2f_{max}$ samples per second are taken for t seconds, then $2f_{max}t = N$, and the first harmonic frequency is $1/t = 2f_{max}/N$.
- The symmetry at harmonic N/2 occurs at the frequency $(N/2)(2f_{max}/N) = \mathbf{f_{max}}$, the maximum frequency present in the signal (the **folding frequency**).
- So it is unnecessary to plot beyond N/2: N real data values transform to N/2 complex DFT values of practical significance.



Key idea لو x(n) حقيقية يبقى X(N − k) هو الـ conjugate بتاع X(k)، فالـ amplitude متماثل حوالين N/2، وده بيقابل الـ folding frequency Fs/2.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.1 (folding at N/2) · p. 19

Q5 Less frequent

Define the Inverse Discrete Fourier Transform (IDFT).

To obtain the discrete transformation from the frequency to the time domain, the IDFT is used:

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT}, \quad n = 0, 1, \ldots, N-1 \quad (2.7)$$

The inverse Fourier transform is obtained from the IDFT by dividing by T: **IFT = IDFT / T**.

Key idea الـ IDFT نفس شكل الـ DFT بالظبط، بس الإشارة في الأُس موجبة (+j) وبنقسم على N.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.2 · p. 21

Q7 Less frequent

State Parseval's theorem.

The normalized energy in the signal is given by either of the expressions:

$$\sum_{n=0}^{N-1} x^2(n) = (1/N) \sum_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

The left-hand side is the sum of the squared magnitudes of the time series; the right-hand side is the mean square spectral amplitude.

Lecture 5

Key idea الطاقة واحدة في الوقت وفي التردد: مجموع x^2 يساوي $(1/N)$ في مجموع $|X(k)|^2$.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.3 · p. 27

★ Cheat sheet · Lecture 5

Remember

Q1 $X(k) = \Sigma\, x(nT)\, e^{-jk\Omega nT}$, $k = 0 \ldots N-1$: N samples in time give N complex frequency components spaced $\Omega = 2\pi/NT$.

Q2 Sampling the Fourier-transform integral gives $F(j\omega) = X(k)\cdot T$, i.e. FT = DFT × T (and IFT = IDFT / T).

Q3 $e^{-j2\pi n} = 1$ for integer n, so $X(k+N) = X(k)$: the DFT repeats every N.

Q4 For real x(n), $X(N-k) = X^*(k)$, so $|X(k)|$ is symmetric about harmonic N/2 (the folding frequency Fs/2).

Q5 $x(nT) = (1/N)\, \Sigma\, X(k)\, e^{jk\Omega nT}$: same form as the DFT with +j and a factor 1/N.

Q6 Symmetry, even and odd functions, Parseval's theorem, delta function, linear cross-correlation and circular convolution.

Q7 $\Sigma\, x^2(n) = (1/N)\, \Sigma\, |X(k)|^2$: the energy is the same in time and in frequency.

Q8 FT → non-periodic continuous-time signals (✓); DFT → sampled sequence x(nT), not the analog signal (× as written).

Equations

$$X(k) = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \quad (2.4)$$

$$X(k) = R(k) + jI(k) \quad (2.1) \qquad |X(k)| = [R^2(k) + I^2(k)]^{\frac{1}{2}} \quad (2.2) \qquad \Phi(k) = \tan^{-1}[I(k)/R(k)] \quad (2.3)$$

$$\sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}\, T = F(j\omega) \quad (2.5)$$

$$\mathbf{F(j\omega) = X(k)\, T} \quad \text{i.e.} \quad \mathbf{FT = DFT \times T} \quad (2.6)$$

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \sum_{k=0}^{N-1} X(k)\, e^{jk\Omega nT}, \quad n = 0, 1, \ldots, N-1 \quad (2.7)$$

$$\sum_{n=0}^{N-1} x^2(n) = (1/N) \sum_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

SECTION 02

Lecture 6

FFT, Butterfly & IFFT

Questions

- ☐ **Q9** Discuss the computational complexity of the DFT. p. 11
- ☐ **Q10** Explain the decimation-in-time (DIT) FFT algorithm. p. 12
- ☐ **Q11** What is the butterfly? Draw it. p. 13
- ☐ **Q12** What are the computational advantages (saving) of the FFT? p. 14
- ☐ **Q13** Show how the IFFT is obtained from the FFT. p. 15

Q9

Discuss the computational complexity of the DFT.

For an 8-point DFT (Eq. 2.15): $X(k) = x(0)e^{-jK0} + x(1)e^{-jK1} + \ldots + x(7)e^{-jK7}$, k = 0, ..., 7.

- Each X(k) needs 8 complex multiplications and 7 complex additions; for N points: N and N − 1.
- There are N harmonic components to evaluate (k = 0 ... N − 1).

Operation	DFT
No. of complex multiplications	N^2
No. of complex additions	$N(N-1)$

If N = 1024, approximately one million complex multiplications and one million complex additions are required, so some means of reducing these numbers is desirable.

Key idea الـ DFT المباشر محتاج N^2 ضرب complex و $N(N-1)$ جمع، والرقم ده بيكبر بسرعة جدًا مع N.

Source — from your PDFs

Lecture **LEC 5** · Sec. 2.4 · p. 30

Q10

Explain the decimation-in-time (DIT) FFT algorithm.

- The FFT uses the computational redundancy in the DFT to reduce the number of calculations (Cooley and Tukey, 1965). Decimation refers to the reduction in calculations performed on time-domain data.
- Writing $W_N = e^{-j2\pi/N}$ (2.17), the DFT becomes $X(k) = \Sigma\, x_n\, W_N^{kn}$ (2.18), with:

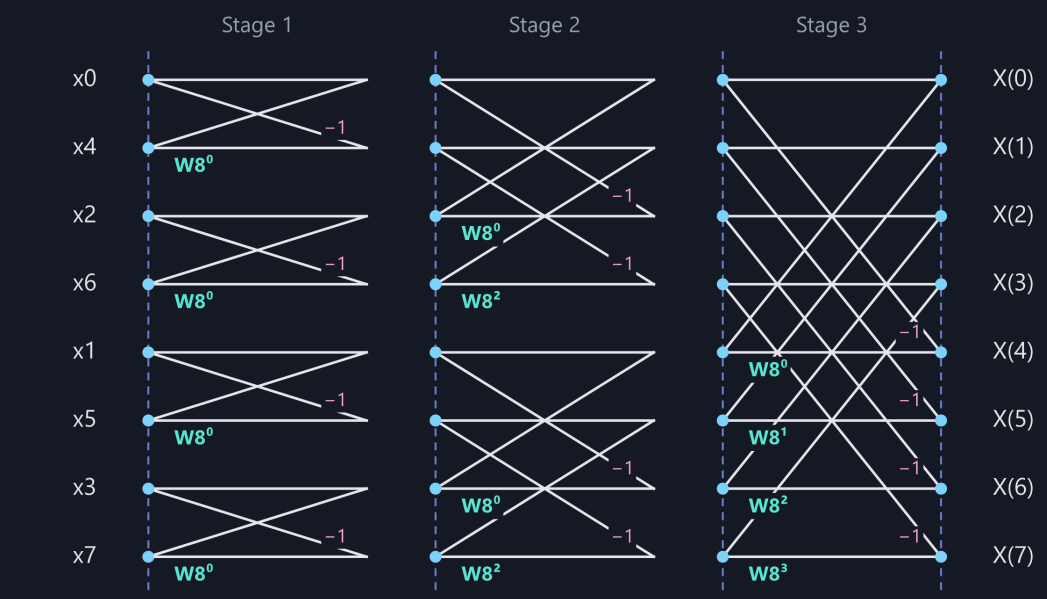
$$W_N^2 = W_{N/2} \quad (2.21b) \qquad W_N^{k+N/2} = -W_N^k \quad (2.21c)$$

- The data are split into the even sequence x_{2n} and the odd sequence x_{2n+1} (N/2 points each):

$$X1(k) = X11(k) + W_N^k\, X12(k) \quad (2.24)$$

- X11(k) is the DFT of the even sequence and X12(k) of the odd sequence. Repeating the decomposition, an 8-point DFT becomes four 2-point DFTs (Table 2.1).

Lecture 6



Key idea بنقسم x(n) لـ even و odd samples ونكرر لحد ما يفضل 2-point DFTs بس (log₂N stage)، وبعدين نجمّعهم بالـ W_N^k.

Source — from your PDFs

Lecture **LEC 6** · Sec. 2.5 · p. 6

Q11

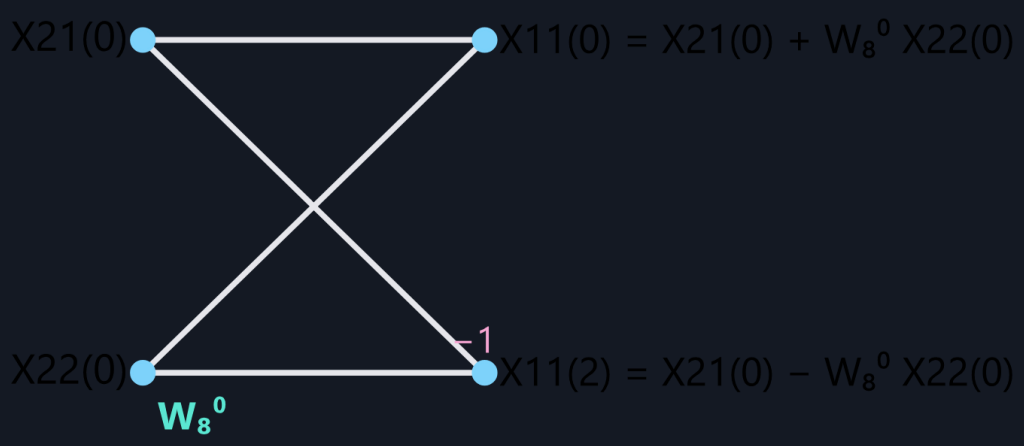
What is the butterfly? Draw it.

Equations 2.25 are represented diagrammatically by taking them in pairs, exploiting the symmetry centred on the sign differences:

$$X11(0) = X21(0) + W_8^0\ X22(0) \qquad X11(2) = X21(0) - W_8^0\ X22(0)$$

The inputs are at the left of the cross and the outputs at the right. The structure is called a **butterfly**, being reminiscent of the symbolic representation of the insect (Fig. 2.2).

The butterfly (Fig. 2.2 a)



Key idea الـ butterfly هو الوحدة اللي بيتبني منها الـ FFT: ضربة واحدة في W وجمعتين بتطلّع خرجين، واحد بـ + والتاني بـ −.

Source — from your PDFs

Lecture **LEC 6** · Sec. 2.5.1 · p. 13

Q12

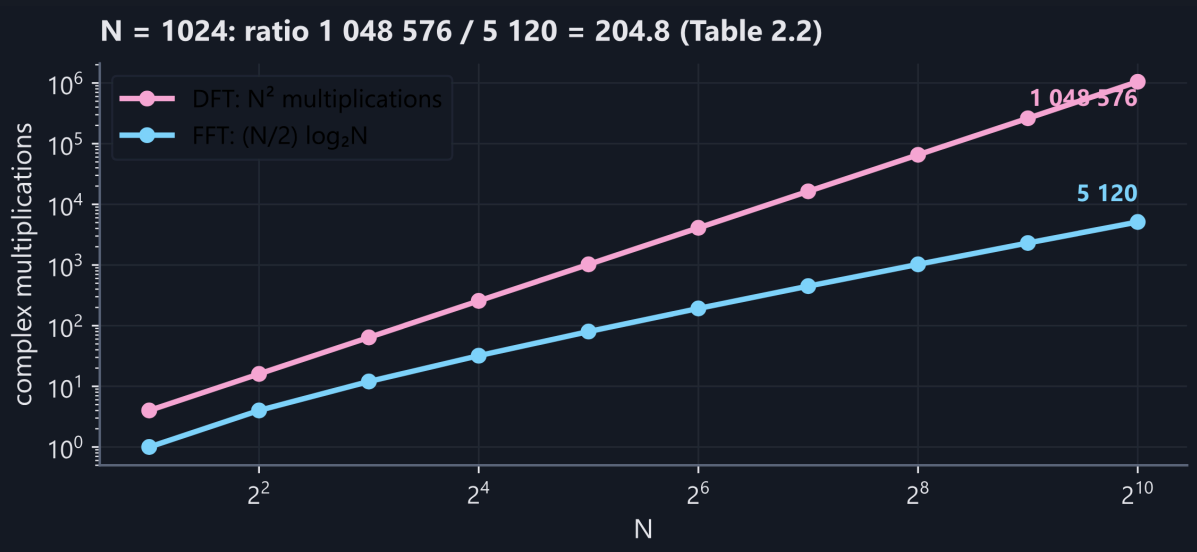
What are the computational advantages (saving) of the FFT?

- An N-point FFT contains N/2 butterflies per stage and $\log_2 N$ stages: $(N/2)\log_2 N$ butterflies.
- Each butterfly involves one complex multiplication and two complex additions.

	DFT	FFT	Saving
Complex multiplications	N^2	$(N/2)\log_2 N$	$N^2 - (N/2)\log_2 N$
Complex additions	$N(N-1)$	$N\log_2 N$	$N(N-1) - N\log_2 N$

N	DFT mult.	DFT add.	FFT mult.	FFT add.
4	16	12	4	8
8	64	56	12	24
1024	1 048 576	1 047 552	5 120	10 240

For N = 1024 the number of multiplications is reduced by a factor of 204.8.



Key idea بنعمل N = 8 مع يعني، N^2 جمع بدل $N \log_2 N$ ضرب و $(N/2)\log_2 N$ محتاج FFT الـ ضربة بدل 12 .64

Source — from your PDFs

Lecture **LEC 6** · Sec. 2.5.2 · p. 18

Q13

Show how the IFFT is obtained from the FFT.

Substituting in Eq. 2.4 (sum over λ, k → μ, Ω = 2π/NT):

$$\text{FFT:} \quad X(\mu) = \Sigma_{\lambda=0}^{N-1} x(\lambda)\, e^{-j\mu(2\pi/N)\lambda} \quad (2.26)$$

and similarly in Eq. 2.7 (k = λ, n = μ):

$$\text{IFFT:} \quad x(\mu) = (1/N) \sum_{\lambda=0}^{N-1} X(\lambda)\, e^{j\mu(2\pi/N)\lambda} \quad (2.27)$$

The IFFT differs from the FFT only in the factor **1/N** and the **sign of the exponent**, so with small modifications the FFT may be used to calculate the IFFT.

Procedure: re-order X(k) as in the FFT, use the same butterflies with W_N^{-k} instead of W_N^{k}, then divide the outputs by N.

Key idea نستخدم نفس الـ flowgraph بتاع الـ FFT، بس بـ W^(−k) بدل W^k، وفي الآخر نقسم الخرج على N.

Key words Procedure

Source — from your PDFs

Lecture **LEC 6** · Sec. 2.6 · p. 20

★ Cheat sheet · Lecture 6

Remember

Q9 Direct DFT needs N^2 complex multiplications (N(N − 1) additions), which grows very fast with N.

Q10 Split x(n) into even and odd samples repeatedly ($\log_2 N$ stages) until only 2-point DFTs remain, then combine with W_N^k.

Q11 Butterfly: X11(0) = X21(0) + W X22(0) and X11(2) = X21(0) − W X22(0) — one multiplication, two additions.

Q12 FFT needs (N/2) $\log_2 N$ multiplications and N $\log_2 N$ additions instead of N^2 — e.g. 12 vs 64 for N = 8.

Q13 Use the same flowgraph with W^{−k} instead of W^{k} and divide the result by N.

Equations

$$W_N^2 = W_{N/2} \quad (2.21b) \qquad W_N^{k+N/2} = -W_N^k \quad (2.21c)$$

$$X1(k) = X11(k) + W_N^k\, X12(k) \quad (2.24)$$

$$\text{FFT:}\quad X(\mu) = \Sigma_{\lambda=0}^{N-1}\, x(\lambda)\, e^{-j\mu(2\pi/N)\lambda} \qquad (2.26)$$

$$\text{IFFT:}\quad x(\mu) = (1/N)\, \Sigma_{\lambda=0}^{N-1}\, X(\lambda)\, e^{j\mu(2\pi/N)\lambda} \qquad (2.27)$$

SECTION 03

Solving Method

The doctor's steps for every problem type

Questions

- ☐ M1 DFT by the direct method (4-point) p. 17
- ☐ M2 IDFT check p. 18
- ☐ M3 FFT by decimation in time (flowgraph) p. 18
- ☐ M4 IFFT, Parseval, spectra, saving and FT p. 19

M1

DFT by the direct method (4-point)

1. Write the definition: **$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}$**, $\Omega = 2\pi/NT$, N = 4 (2.4).

2. List the samples: x(0), x(T), x(2T), x(3T). It is required to find X(k) for k = 0, 1, 2, 3.

3. For every k write the four terms, then expand each exponential: **$e^{-j\theta} = \cos\theta - j\sin\theta$** ($e^{-j\pi/2} = -j$, $e^{-j\pi} = -1$, $e^{-j3\pi/2} = +j$).

4. Add the terms → X(k). Write "of magnitude $|X(k)|$ and phase angle $\Phi(k)$" after each one.

5. Collect the answer: X(k) = {X(0), X(1), X(2), X(3)}.

DOCTOR'S STYLE

Examples: Example 2.1 · Sheet 2.2 · P9 to P13

Solving Method

M2

IDFT check

1. Definition: **$x(nT) = (1/N) \Sigma_{k=0}^{N-1} X(k) e^{+j2\pi kn/N}$** (2.7): the same as the DFT with +j and a factor 1/N.

2. For every n = 0 ... 3 write the four terms, expand $e^{+j\theta} = \cos\theta + j \sin\theta$, add and multiply by ¼.

3. End each line with "= x(n), as expected" and write the original sequence at the end.

DOCTOR'S STYLE
Examples: Example 2.2 · Sheet 2.2

M3

FFT by decimation in time (flowgraph)

1. Re-order the input by bit reversal: N = 4 → **x0, x2, x1, x3**; N = 8 → **x0, x4, x2, x6, x1, x5, x3, x7**.

2. First stage, 2-point butterflies (W = 1): X21(0) = x0 + x2, X21(1) = x0 − x2, X22(0) = x1 + x3, X22(1) = x1 − x3.

3. Next stage: **$X11(k) = X21(k) + W_N^k X22(k)$** and **$X11(k + N/2) = X21(k) - W_N^k X22(k)$** (2.24).

4. W values: $W_8^0 = 1$, $W_8^1 = 0.707 - j0.707$, $W_8^2 = -j$, $W_8^3 = -0.707 - j0.707$. On the flowgraph, $W = W_M^k$ with $M = 2^{stage}$.

5. Draw the flowgraph (Fig. 2.3) with the values on it, and say the results are the same as those obtained by the DFT.

DOCTOR'S STYLE
Examples: Example 2.3 · Example 2.4 · Sheet 2.4, 2.5, 2.6 · Assignment

M4

IFFT, Parseval, spectra, saving and FT

1. IFFT: re-order X(k) the same way, use the same flowgraph with W^{-k} ($W_8^{-2} = +j$), then divide the outputs by N (Sec. 2.6).

2. Parseval: $\Sigma\, x^2(n) = (1/N)\, \Sigma\, |X(k)|^2$ (2.12). Compute L.H.S. and R.H.S. and write "L.H.S. = R.H.S.".

3. Spectra: table of k, X(k), $|X(k)|$, $\Phi(k)$, then two stem plots (amplitude and phase).

4. Saving: DFT needs N^2 multiplications and $N(N-1)$ additions; FFT needs $(N/2)\log_2 N$ multiplications and $N \log_2 N$ additions; saving = the difference.

5. Fourier transform: FT = T × DFT (Example 2.4, T = 1/Fs).

DOCTOR'S STYLE

Examples: Example 2.2 · Example 2.4 · Sheet 2.3 to 2.6 · P9 to P13

Key words IFFT Parseval Spectra Saving

SECTION 04

Lecture Examples

Examples 2.1 to 2.4, solved step by step

Questions

- ☐ P1 Example 2.1 — DFT of {1, 0, 0, 1} p. 20
- ☐ P2 Example 2.2 — IDFT of {2, 1 + j, 0, 1 − j} p. 22
- ☐ P3 Example 2.3 — FFT of {1, 0, 0, 1} p. 23
- ☐ P4 Example 2.4 — FFT of {0.5, 1, 1, 0.5}, sampled at 125 Hz p. 25

Lecture Examples

P1

Example 2.1 — DFT of {1, 0, 0, 1}

The DFT of the sequence {1, 0, 0, 1} (sampled at 8 kHz) will be evaluated. Find X(k), plot |X(k)| and Φ(k), and find the Fourier transform components.

$$X(k) = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \sum_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 0 + 1 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi/2} \\ &= 1 + [\cos(3\pi/2) - j\, \sin(3\pi/2)] \\ &= 1 + j \\ &= \mathbf{1 + j} \end{aligned}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$\begin{aligned} X(2) &= \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + 0 + e^{-j3\pi} \\ &= 1 + [\cos(3\pi) - j\, \sin(3\pi)] \\ &= 1 + (-1) \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(9\pi/2) - j \sin(9\pi/2)]$$
$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

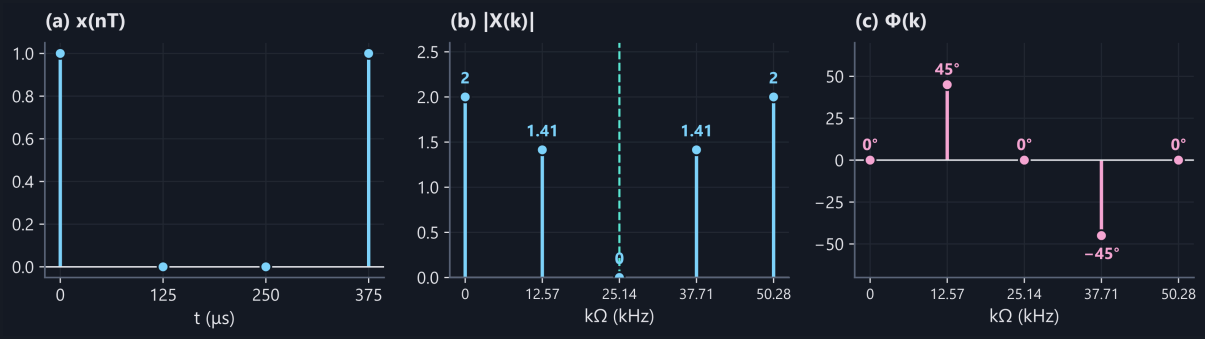
Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
|X(k)|	2	√2 = 1.414	0	√2 = 1.414
Φ(k)	0°	45°	0°	−45°

$T = 1/(8\times10^{3}) = 125\ \mu s \qquad \Omega = 2\pi/NT = 2\pi/(4 \times 125\times10^{-6}) = 12.57\ k \rightarrow 2\Omega = 25.14\ k,\ 3\Omega = 37.71\ k$

The Fourier transform components are obtained by multiplying the DFT components by T = 125 μs (FT = DFT × T):

$F(0) = 250\ \mu V\ Hz^{-1}, \quad F(\Omega) = (125 + j125)\ \mu V\ Hz^{-1}, \quad F(2\Omega) = 0, \quad F(3\Omega) = (125 - j125)\ \mu V\ Hz^{-1}$



Lecture Examples

Result X(k) = {2, 1 + j, 0, 1 − j} · |X(k)| = {2, 1.414, 0, 1.414} · Φ(k) = {0°, 45°, 0°, −45°}

Key idea نعوّض في معادلة الـ DFT لكل k من 0 لـ 3، ونفك كل e^(−jθ) لـ cos − j sin، ونجمع. وبعدين نطلع الـ magnitude والـ phase لكل X(k).

Source — from your PDFs

Lecture **LEC 5** · Example 2.1 · p. 11

P2

Example 2.2 — IDFT of {2, 1 + j, 0, 1 − j}

Derive the time series {1, 0, 0, 1} from its DFT components {2, 1 + j, 0, 1 − j}.

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j) + 0 + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi/2} + 0 + (1 - j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= \tfrac{1}{4}\ [2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi} + 0 + (1 - j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j3\pi/2} + 0 + (1 - j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-j) + (1 - j)(j)]$$
$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 0, 1}**, the original sequence.

Result x(n) = {1, 0, 0, 1} (the original sequence)

Key idea نعوّض في معادلة الـ IDFT لكل n من 0 لـ 3، وبالإشارة الموجبة وبعد القسمة على 4 بنرجع للـ sequence الأصلية.

Source — from your PDFs

Lecture **LEC 5** · Example 2.2 · p. 23

P3

Example 2.3 — FFT of {1, 0, 0, 1}

Obtain the DFT of {1, 0, 0, 1} by means of the decimation-in-time FFT algorithm.

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

$$X1(k) = X11(k) + W_N^k\ X12(k), \quad k = 0, 1, 2, 3 \ (2.24)$$

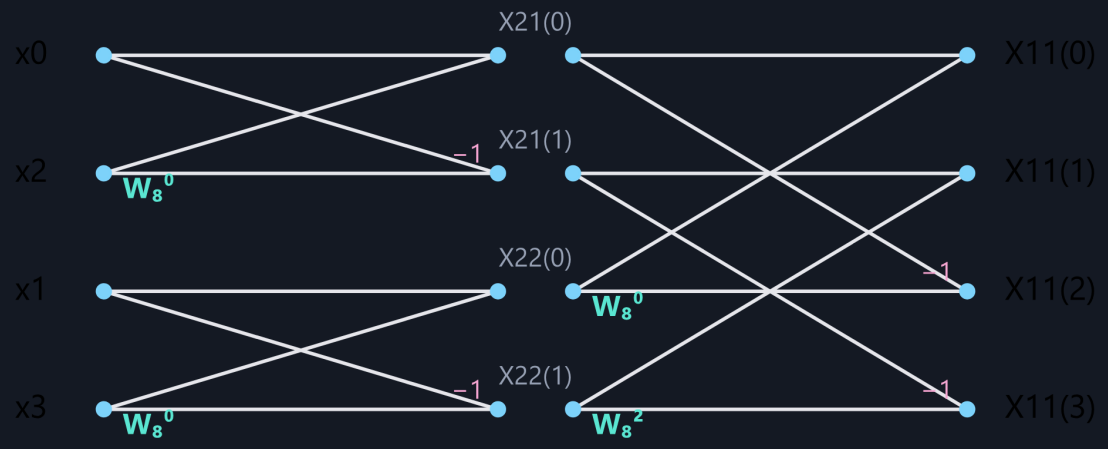
The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

Lecture Examples

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained in Section 2.1, but were more readily obtained using the FFT algorithm.

Result X(k) = {2, 1 + j, 0, 1 − j} — same as the DFT of Example 2.1

Key idea نفس الـ sequence بس بالـ FFT: نرتب الـ input x0, x2, x1, x3، ونعمل 2-point butterflies، وبعدين نجمّع بالـ W، فبيطلع نفس ناتج الـ DFT بحسابات أقل.

Source — from your PDFs

Lecture **LEC 6** · Example 2.3 · p. 16

P4

Example 2.4 — FFT of {0.5, 1, 1, 0.5}, sampled at 125 Hz

The first four sampled voltage values of a 10 Hz bandwidth signal sampled at 125 Hz were (0.5, 1, 1, 0.5). Obtain the DFT using the FFT, and hence the Fourier transform.

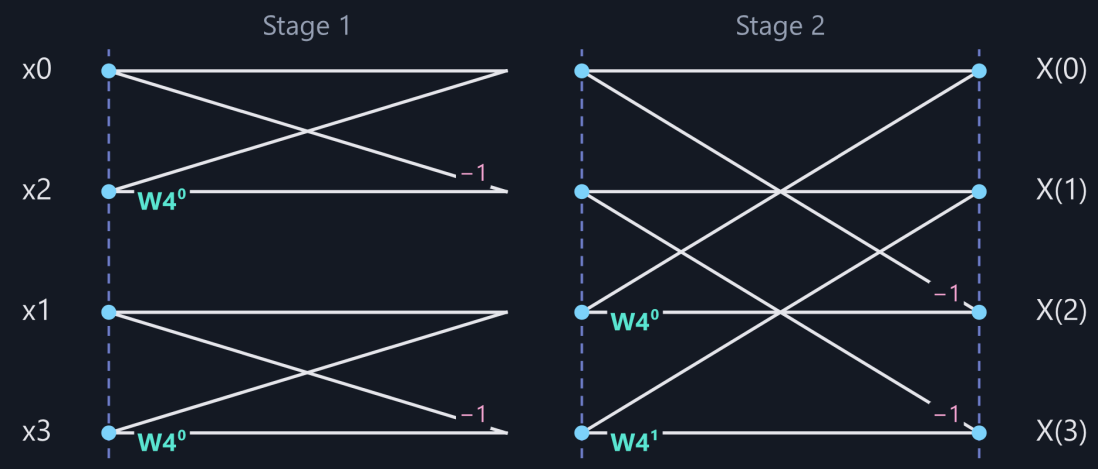
Re-ordered sequence: x(0), x(2), x(1), x(3). $W_4^0 = 1$, $W_4^1 = e^{-j2\pi/4} = -j$

$$G(0) = x(0) + x(2) = 1.5 \qquad G(1) = x(0) - x(2) = -0.5$$
$$H(0) = x(1) + x(3) = 1.5 \qquad H(1) = x(1) - x(3) = 0.5$$

$$X(0) = G(0) + W_4^0 H(0) = 1.5 + 1.5 = \mathbf{3}$$
$$X(1) = G(1) + W_4^1 H(1) = -0.5 - j(0.5) = \mathbf{-0.5 - j0.5}$$
$$X(2) = G(0) - W_4^0 H(0) = 1.5 - 1.5 = \mathbf{0}$$
$$X(3) = G(1) - W_4^1 H(1) = -0.5 + j(0.5) = \mathbf{-0.5 + j0.5}$$

∴ X(k) = **{3, −0.5 − j0.5, 0, −0.5 + j0.5}** (same as the DFT).

4-point DIT FFT flowgraph (Fig. 2.4)



T = 1/125 = 0.008 s; the signal period is 1/10 = 0.1 s → T/period = 0.08 ≪ 1

→ FT = T × DFT

FT = {0.024, −0.004(1 + j), 0, 0.004(−1 + j)}

Result $X(k) = \{3, -0.5 - j0.5, 0, -0.5 + j0.5\} \cdot FT = \{0.024, -0.004(1 + j), 0, 0.004(-1 + j)\}$

Key idea FFT عادي لـ 4 samples، وبعدين نضرب في $T = 1/125$ عشان نطلع الـ Fourier transform الحقيقي (FT = T × DFT).

Source — from your PDFs

Lecture **LEC 6** · Example 2.4 · p. 22

★ Cheat sheet · Lecture Examples

Final results

P1	$X(k) = \{2, 1 + j, 0, 1 - j\} \cdot \lvert X(k)\rvert = \{2, 1.414, 0, 1.414\} \cdot \Phi(k) = \{0°, 45°, 0°, -45°\}$
P2	$x(n) = \{1, 0, 0, 1\}$ (the original sequence)
P3	$X(k) = \{2, 1 + j, 0, 1 - j\}$ — same as the DFT of Example 2.1
P4	$X(k) = \{3, -0.5 - j0.5, 0, -0.5 + j0.5\} \cdot FT = \{0.024, -0.004(1 + j), 0, 0.004(-1 + j)\}$

SECTION 05

Previous Exams

Midterms, finals & Revision Sheet

Questions

☐	**P9**	$x(n) = \{1, 0, 1, 1\}$	p. 27
☐	**P10**	$x(n) = \{1, 0, 1, 0\}$	p. 32
☐	**P11**	$x(n) = \{1, 1, 0, 1\}$	p. 37
☐	**P12**	$x(n) = \{1, 0, 0, 1\}$	p. 42
☐	**P13**	$x(n) = \{1, 1, 1, 0\}$	p. 47

P9 ● Frequent 10×

x(n) = {1, 0, 1, 1}

For the sequence x(n) = {1, 0, 1, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \text{ (2.4)}$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 1 + 1 \\ &= \mathbf{3} \end{aligned}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 1 + 0 + e^{-j\pi} + e^{-j3\pi/2} \\ &= 1 + [\cos(\pi) - j\sin(\pi)] + [\cos(3\pi/2) - j\sin(3\pi/2)] \\ &= 1 + (-1) + j \\ &= \mathbf{j} \end{aligned}$$

Thus X(1) = j, of magnitude 1 and phase angle Φ(1) = 90°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 1 + 0 + e^{-j2\pi} + e^{-j3\pi} \\ &= 1 + [\cos(2\pi) - j\sin(2\pi)] + [\cos(3\pi) - j\sin(3\pi)] \\ &= 1 + 1 + (-1) \\ &= \mathbf{1} \end{aligned}$$

Thus X(2) = 1, of magnitude 1 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \sum_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi) - j\sin(3\pi)] + [\cos(9\pi/2) - j\sin(9\pi/2)]$$
$$= 1 + (-1) + (-j)$$
$$= \mathbf{-j}$$

Thus X(3) = –j, of magnitude 1 and phase angle Φ(3) = −90°.

So, the time series x(n) = {1, 0, 1, 1} has the DFT given by **X(k) = {3, j, 1, –j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 1, and

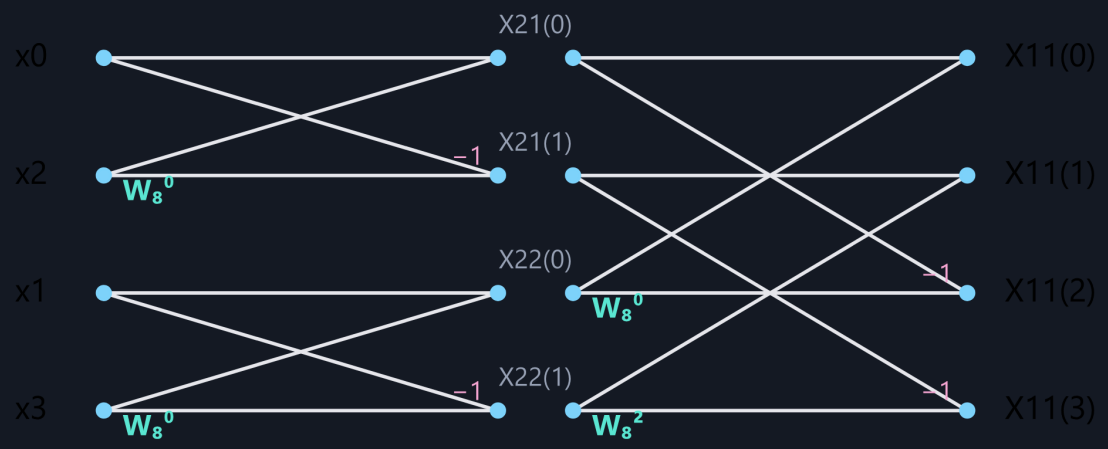
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 2 + 1 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 0 + e^{-j\pi/2}(-1) = 0 - j(-1) = \mathbf{j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 2 - 1 = \mathbf{1}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 0 - e^{-j\pi/2}(-1) = 0 + j(-1) = \mathbf{-j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + jj2] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - jj2] = \mathbf{1}$$

∴ x(n) = **{1, 0, 1, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j\pi/2} + 1\ e^{j\pi} + (-j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(j) + 1(-1) + (-j)(-j)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j\pi} + 1\ e^{j2\pi} + (-j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + j(-1) + 1(1) + (-j)(-1)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + j\ e^{j3\pi/2} + 1\ e^{j3\pi} + (-j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + j(-j) + 1(-1) + (-j)(j)]$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 1, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)|$ = {3, 1, 1, 1}

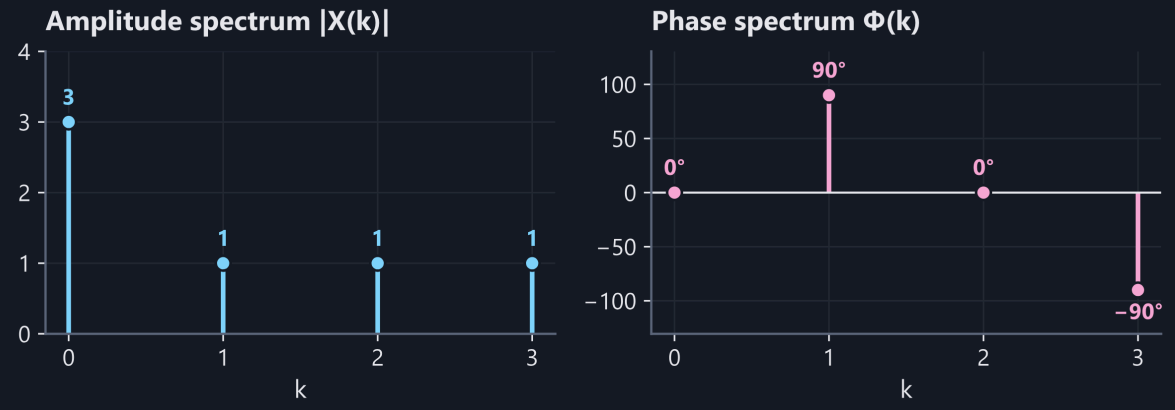
$$L.H.S. = 1^2 + 0^2 + 1^2 + 1^2 = \mathbf{3}$$
$$R.H.S. = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	j	1	−j
|X(k)|	3	1	1	1
Φ(k)	0°	90°	0°	−90°

Spectra of {1, 0, 1, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Result `X(k) = {3, j, 1, −j} · IDFT → {1, 0, 1, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.`

Key idea أكتر مسألة متكررة في الشابتر: DFT أو FFT، وبعدها check بالـ IDFT أو IFFT، وبعدها Parseval، والـ spectra، والـ saving. نفس الخطوات في كل امتحان.

Key words (iii) Parseval's theorem

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 20-1-2025 · Q4-c · Final 23-1-2023 · Q3-b · Final 17-8-2026 · Q2-b · Final 18-1-2024 · Q2-c · Final 12-1-2019 · Q2-c · Final 2-1-2017 · Q2-c · Midterm 31-7-2023 · Q2-b · Midterm 8-12-2020 · Q2-c · Midterm 2 · Q2-c · Book Assignment A2.2

P10 · Frequent · 4×

x(n) = {1, 0, 1, 0}

For the sequence x(n) = {1, 0, 1, 0}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 1 + 0 + 1 + 0 \\ &= 2 \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + 0 + e^{-j\pi} + 0$$
$$= 1 + [\cos(\pi) - j\,\sin(\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(1) = 0, of magnitude 0 and phase angle Φ(1) = 0°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + 0 + e^{-j2\pi} + 0$$
$$= 1 + [\cos(2\pi) - j\,\sin(2\pi)]$$
$$= 1 + 1$$
$$= \mathbf{2}$$

Thus X(2) = 2, of magnitude 2 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + e^{-j3\pi} + 0$$
$$= 1 + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(3) = 0, of magnitude 0 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 0, 1, 0} has the DFT given by **X(k) = {2, 0, 2, 0}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 1, x3 = 0, and

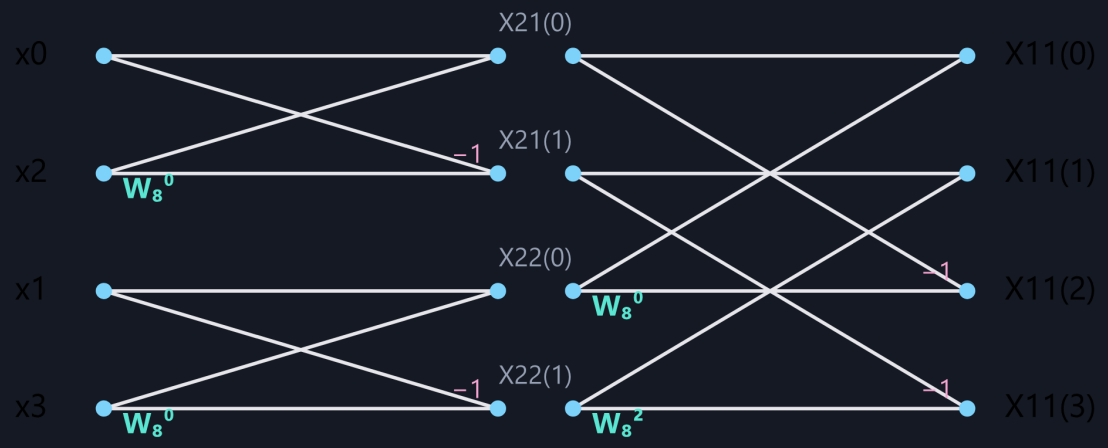
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 0 + 0 = 0$$
$$X22(1) = x1 - x3 = 0 - 0 = 0$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 2 + 0 = 2$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 0 + e^{-j\pi/2}(0) = 0 - j(0) = 0$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 2 - 0 = 2$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 0 - e^{-j\pi/2}(0) = 0 + j(0) = 0$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 0$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = 0$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[0 + j0] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[0 - j0] = \mathbf{0}$$

∴ x(n) = **{1, 0, 1, 0}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2 + 0]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-2)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j2\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(1)]$$
$$= \tfrac{1}{4}\ [2 + 2]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + 0 + 2\ e^{j3\pi} + 0]$$
$$= \tfrac{1}{4}\ [2 + 2(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-2)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 1, 0}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1} x^2(n) = (1/N) \Sigma_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, 0, 2, 0\}$

$$L.H.S. = 1^2 + 0^2 + 1^2 + 0^2 = 2$$
$$R.H.S. = (1/4)[2^2 + 0^2 + 2^2 + 0^2] = (1/4)(8) = 2$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	0	2	0
$\|X(k)\|$	2	0	2	0
Φ(k)	0°	0°	0°	0°

Spectra of {1, 0, 1, 0}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2) \log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N \log_2 N = 8$	**4**

Result X(k) = {2, 0, 2, 0} · IDFT → {1, 0, 1, 0} · Parseval 2 = 2 · saving 12 mult., 4 add.

Key idea نفس الخطوات. الـ X(k) هنا كلها أرقام حقيقية {2, 0, 2, 0}، لأن الـ sequence متكررة كل 2 samples.

Key words (iii) Parseval's theorem

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 11-6-2023 · Q1-c; Final 7-6-2023 · Q1-c; Midterm 12-11-2025 · Q2-b; Midterm 28-11-2021 · Q2-b

P11 2×

x(n) = {1, 1, 0, 1}

For the sequence x(n) = {1, 1, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 1, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 1 + 1 + 0 + 1$$
$$= \mathbf{3}$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + e^{-j\pi/2} + 0 + e^{-j3\pi/2}$$
$$= 1 + [\cos(\pi/2) - j\,\sin(\pi/2)] + [\cos(3\pi/2) - j\,\sin(3\pi/2)]$$
$$= 1 + (-j) + j$$
$$= \mathbf{1}$$

Thus X(1) = 1, of magnitude 1 and phase angle Φ(1) = 0°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + e^{-j\pi} + 0 + e^{-j3\pi}$$
$$= 1 + [\cos(\pi) - j\,\sin(\pi)] + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + (-1) + (-1)$$
$$= \mathbf{-1}$$

Thus X(2) = −1, of magnitude 1 and phase angle Φ(2) = 180°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + e^{-j3\pi/2} + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(3\pi/2) - j\,\sin(3\pi/2)] + [\cos(9\pi/2) - j\,\sin(9\pi/2)]$$
$$= 1 + j + (-j)$$
$$= \mathbf{1}$$

Thus X(3) = 1, of magnitude 1 and phase angle Φ(3) = 0°.

So, the time series x(n) = {1, 1, 0, 1} has the DFT given by **X(k) = {3, 1, −1, 1}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 1, x2 = 0, x3 = 1, and

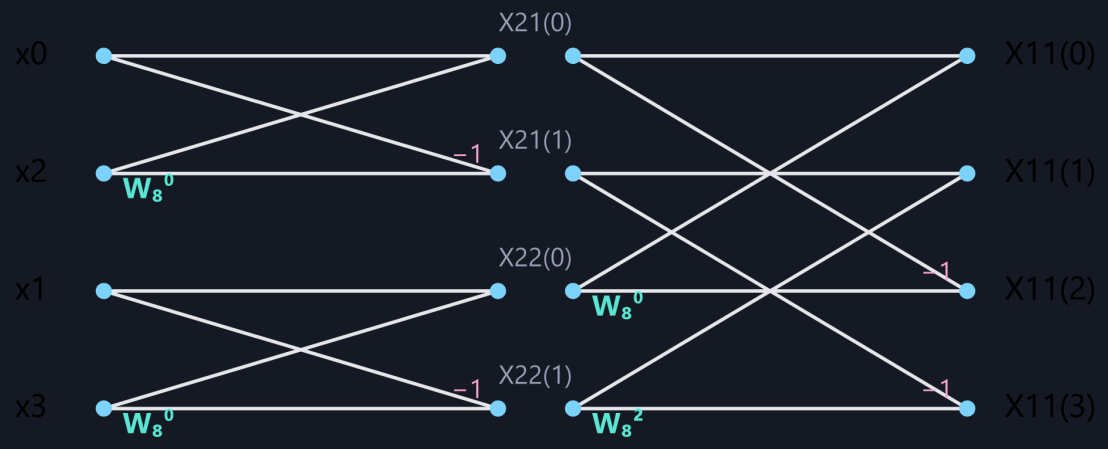
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$\begin{aligned} X21(0) &= x0 + x2 = 1 + 0 = 1 \\ X21(1) &= x0 - x2 = 1 - 0 = 1 \\ X22(0) &= x1 + x3 = 1 + 1 = 2 \\ X22(1) &= x1 - x3 = 1 - 1 = 0 \end{aligned}$$

$$\begin{aligned} X11(0) &= X21(0) + W_8^0\, X22(0) = 1 + 2 = \mathbf{3} \\ X11(1) &= X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(0) = 1 - j(0) = \mathbf{1} \\ X11(2) &= X21(0) - W_8^0\, X22(0) = 1 - 2 = \mathbf{-1} \\ X11(3) &= X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(0) = 1 + j(0) = \mathbf{1} \end{aligned}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$\begin{aligned} x21(0) &= X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 4 \\ x22(0) &= X(1) + X(3) = 2 \qquad x22(1) = X(1) - X(3) = 0 \end{aligned}$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[4 + j0] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[4 - j0] = \mathbf{1}$$

∴ x(n) = **{1, 1, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j\pi/2} + (-1)\ e^{j\pi} + 1\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(j) + (-1)(-1) + 1(-j)]$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j\pi} + (-1)\ e^{j2\pi} + 1\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + 1(-1) + (-1)(1) + 1(-1)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + 1\ e^{j3\pi/2} + (-1)\ e^{j3\pi} + 1\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + 1(-j) + (-1)(-1) + 1(j)]$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 1, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1} x^2(n) = (1/N) \Sigma_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{3, 1, 1, 1\}$

$$\text{L.H.S.} = 1^2 + 1^2 + 0^2 + 1^2 = \mathbf{3}$$
$$\text{R.H.S.} = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	1	−1	1
$\|X(k)\|$	3	1	1	1
Φ(k)	0°	0°	180°	0°

Spectra of {1, 1, 0, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2) \log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N \log_2 N = 8$	**4**

Previous Exams

Result X(k) = {3, 1, −1, 1} · IDFT → {1, 1, 0, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.

Key idea X(k) = {3, 1, −1, 1} حقيقية، لأن الـ sequence متماثلة حوالين n = 0. نفس الخطوات.

Key words (iii) Parseval's theorem

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 9-6-2026 · Q2-b · Midterm 1 Model-2 · Q2-c

P12 3×

x(n) = {1, 0, 0, 1}

For the sequence x(n) = {1, 0, 0, 1}: (i) find the frequency components X(k) using the DFT (or FFT); (ii) check by calculating x(n) using the IFFT (or IDFT); (iii) verify Parseval's theorem; (iv) plot the amplitude and phase spectra; (v) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 0, x(2T) = 0, x(3T) = 1. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 1 + 0 + 0 + 1$$
$$= 2$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + 0 + 0 + e^{-j3\pi/2}$$
$$= 1 + [\cos(3\pi/2) - j\, \sin(3\pi/2)]$$
$$= 1 + j$$
$$= \mathbf{1 + j}$$

Thus X(1) = 1 + j, of magnitude √2 = 1.414 and phase angle Φ(1) = 45°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + 0 + 0 + e^{-j3\pi}$$
$$= 1 + [\cos(3\pi) - j\, \sin(3\pi)]$$
$$= 1 + (-1)$$
$$= \mathbf{0}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + 0 + 0 + e^{-j9\pi/2}$$
$$= 1 + [\cos(9\pi/2) - j\, \sin(9\pi/2)]$$
$$= 1 + (-j)$$
$$= \mathbf{1 - j}$$

Thus X(3) = 1 − j, of magnitude √2 = 1.414 and phase angle Φ(3) = −45°.

So, the time series x(n) = {1, 0, 0, 1} has the DFT given by **X(k) = {2, 1 + j, 0, 1 − j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 0, x2 = 0, x3 = 1, and

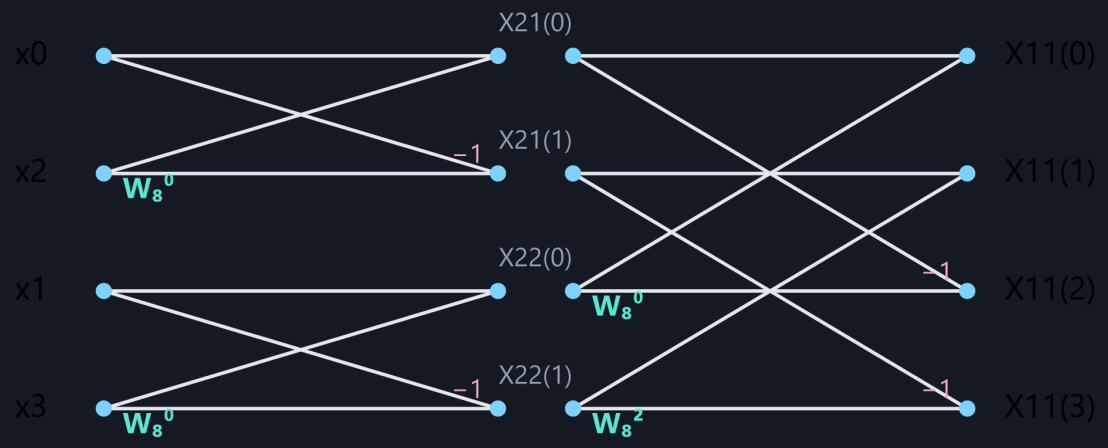
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 0 = 1$$
$$X21(1) = x0 - x2 = 1 - 0 = 1$$
$$X22(0) = x1 + x3 = 0 + 1 = 1$$
$$X22(1) = x1 - x3 = 0 - 1 = -1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = 1 + e^{-j\pi/2}(-1) = 1 - j(-1) = \mathbf{1 + j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = 1 - e^{-j\pi/2}(-1) = 1 + j(-1) = \mathbf{1 - j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 2 \qquad x22(1) = X(1) - X(3) = j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + jj2] = \mathbf{0}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - jj2] = \mathbf{1}$$

∴ x(n) = **{1, 0, 0, 1}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j) + 0 + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi/2} + 0 + (1 - j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(j) + (1 - j)(-j)]$$
$$= \tfrac{1}{4}\ [2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j\pi} + 0 + (1 - j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-1) + (1 - j)(-1)]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j) + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + (1 + j)\ e^{j3\pi/2} + 0 + (1 - j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (1 + j)(-j) + (1 - j)(j)]$$
$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

∴ x(n) = **{1, 0, 0, 1}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1} x^2(n) = (1/N) \Sigma_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$

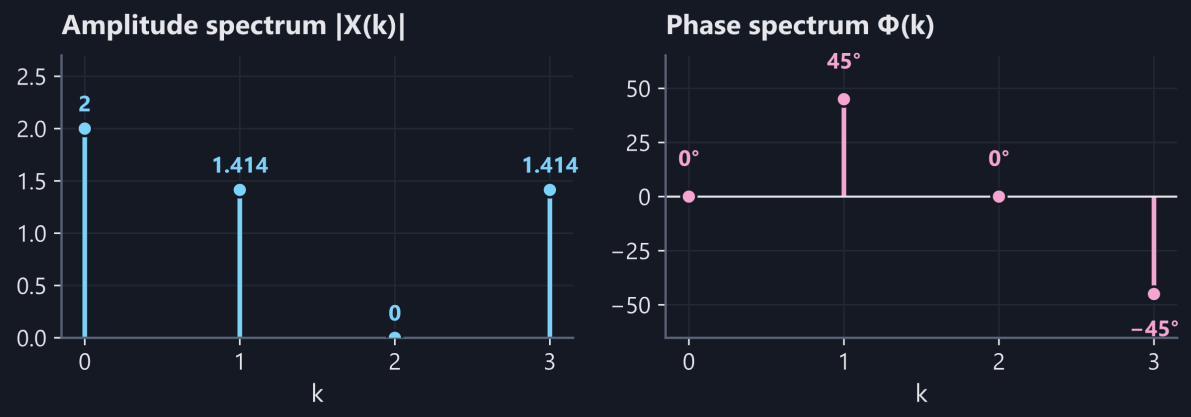
$$L.H.S. = 1^2 + 0^2 + 0^2 + 1^2 = 2$$
$$R.H.S. = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = 2$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	2	1 + j	0	1 − j
$\|X(k)\|$	2	$\sqrt{2}$ = 1.414	0	$\sqrt{2}$ = 1.414
Φ(k)	0°	45°	0°	−45°

Spectra of {1, 0, 0, 1}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	N^2 = 16	$(N/2) \log_2 N$ = 4	**12**
No. of complex additions	$N(N-1)$ = 12	$N \log_2 N$ = 8	**4**

Result X(k) = {2, 1 + j, 0, 1 − j} · IDFT → {1, 0, 0, 1} · Parseval 2 = 2 · saving 12 mult., 4 add.

Key idea دي نفس sequence مثال 2.1، فالـ X(k) = {2, 1 + j, 0, 1 − j}، وبنكمّل بالـ IFFT والـ Parseval والـ spectra.

Key words (iii) Parseval's theorem

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 5** · Sec. 2.3 · p. 27
Lecture	**LEC 6** · Example 2.3 · p. 16
Came in	Final 23-12-2017 · Q2-c · Final 28-12-2019 · Q2-b · Midterm 26-7-2026 · Q2-b

P13 1×

x(n) = {1, 1, 1, 0}

For the sequence x(n) = {1, 1, 1, 0}: (a) using the direct method (DFT) find X(k) and check by the IDFT; (b) repeat using the FFT and check by the IFFT; (c) verify Parseval's theorem; (d) plot the amplitude and phase spectra; (e) find the computational saving of the FFT.

(i) DFT method (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 1, x(T) = 1, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$X(0) = \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4}$$
$$= x(0) + x(T) + x(2T) + x(3T)$$
$$= 1 + 1 + 1 + 0$$
$$= 3$$

Thus X(0) = 3, of magnitude 3 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 1 + e^{-j\pi/2} + e^{-j\pi} + 0$$
$$= 1 + [\cos(\pi/2) - j\,\sin(\pi/2)] + [\cos(\pi) - j\,\sin(\pi)]$$
$$= 1 + (-j) + (-1)$$
$$= \mathbf{-j}$$

Thus X(1) = –j, of magnitude 1 and phase angle Φ(1) = −90°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 1 + e^{-j\pi} + e^{-j2\pi} + 0$$
$$= 1 + [\cos(\pi) - j\,\sin(\pi)] + [\cos(2\pi) - j\,\sin(2\pi)]$$
$$= 1 + (-1) + 1$$
$$= \mathbf{1}$$

Thus X(2) = 1, of magnitude 1 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 1 + e^{-j3\pi/2} + e^{-j3\pi} + 0$$
$$= 1 + [\cos(3\pi/2) - j\,\sin(3\pi/2)] + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= 1 + j + (-1)$$
$$= \mathbf{j}$$

Thus X(3) = j, of magnitude 1 and phase angle Φ(3) = 90°.

So, the time series x(n) = {1, 1, 1, 0} has the DFT given by **X(k) = {3, –j, 1, j}**.

FFT algorithm (as Example 2.3):

This is a four-point DFT with x0 = 1, x1 = 1, x2 = 1, x3 = 0, and

$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 1 + 1 = 2$$
$$X21(1) = x0 - x2 = 1 - 1 = 0$$
$$X22(0) = x1 + x3 = 1 + 0 = 1$$
$$X22(1) = x1 - x3 = 1 - 0 = 1$$

$$X11(0) = X21(0) + W_8^0\ X22(0) = 2 + 1 = \mathbf{3}$$
$$X11(1) = X21(1) + W_8^2\ X22(1) = 0 + e^{-j\pi/2}(1) = 0 - j(1) = \mathbf{-j}$$
$$X11(2) = X21(0) - W_8^0\ X22(0) = 2 - 1 = \mathbf{1}$$
$$X11(3) = X21(1) - W_8^2\ X22(1) = 0 - e^{-j\pi/2}(1) = 0 + j(1) = \mathbf{j}$$

[[s2_bf4]]

These values are the same as those obtained by the DFT.

(ii) Check by the IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 4 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = 0 \qquad x22(1) = X(1) - X(3) = -j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + j(-j2)] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - j(-j2)] = \mathbf{0}$$

$\therefore$ x(n) = **{1, 1, 1, 0}**, the original data sequence.

Check by the IDFT (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{jk\Omega nT} = (1/N)\ \Sigma_{k=0}^{N-1}\ X(k)\ e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\ [3 + (-j) + 1 + j]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [3 + (-j)\ e^{j\pi/2} + 1\ e^{j\pi} + j\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + (-j)(j) + 1(-1) + j(-j)]$$
$$= \tfrac{1}{4}\ [3 + 1 + (-1) + 1]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [3 + (-j)\ e^{j\pi} + 1\ e^{j2\pi} + j\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [3 + (-j)(-1) + 1(1) + j(-1)]$$
$$= \tfrac{1}{4}\ [3 + j + 1 + (-j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [3 + (-j)\ e^{j3\pi/2} + 1\ e^{j3\pi} + j\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [3 + (-j)(-j) + 1(-1) + j(j)]$$
$$= \tfrac{1}{4}\ [3 + (-1) + (-1) + (-1)]$$
$$= \mathbf{0}, \text{ as expected.}$$

$\therefore$ x(n) = **{1, 1, 1, 0}**, the original sequence.

(iii) Parseval's theorem:

$$\Sigma_{n=0}^{N-1}\ x^2(n) = (1/N)\ \Sigma_{k=0}^{N-1}\ |X(k)|^2 \quad (2.12)$$

$|X(k)|$ = {3, 1, 1, 1}

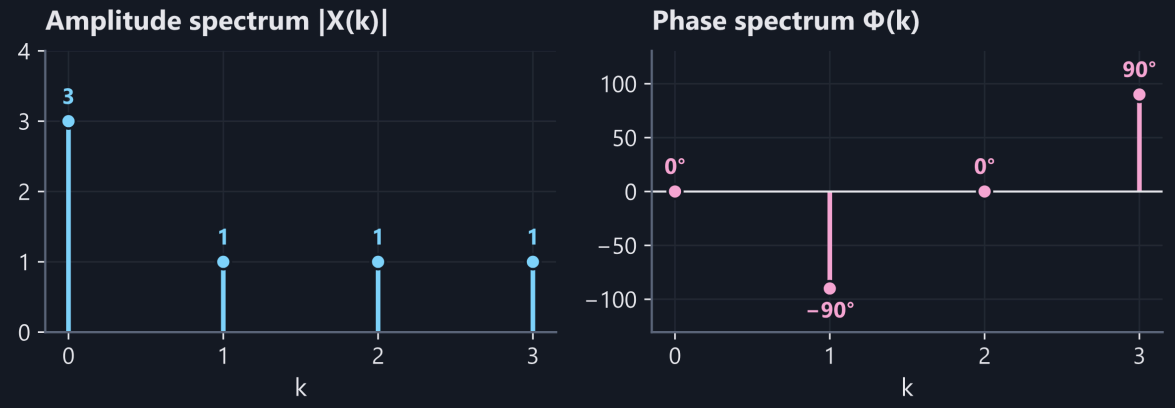
$$L.H.S. = 1^2 + 1^2 + 1^2 + 0^2 = \mathbf{3}$$
$$R.H.S. = (1/4)[3^2 + 1^2 + 1^2 + 1^2] = (1/4)(12) = \mathbf{3}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

(iv) Amplitude and phase spectra:

k	0	1	2	3
X(k)	3	−j	1	j
|X(k)|	3	1	1	1
Φ(k)	0°	−90°	0°	90°

Spectra of {1, 1, 1, 0}



(v) Computational saving (N = 4):

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Result X(k) = {3, −j, 1, j} · IDFT → {1, 1, 1, 0} · Parseval 3 = 3 · saving 12 mult., 4 add.

Key idea نفس الخطوات على {1, 1, 1, 0}: الـ X(k) = {3, −j, 1, j}، والـ Parseval بيطلع 3 = 3.

Key words (iii) Parseval's theorem

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Came in Revision Sheet 1 · Ch 2, Q2

★ Cheat sheet · Previous Exams

Final results

P9	X(k) = {3, j, 1, −j} · IDFT → {1, 0, 1, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.
P10	X(k) = {2, 0, 2, 0} · IDFT → {1, 0, 1, 0} · Parseval 2 = 2 · saving 12 mult., 4 add.
P11	X(k) = {3, 1, −1, 1} · IDFT → {1, 1, 0, 1} · Parseval 3 = 3 · saving 12 mult., 4 add.
P12	X(k) = {2, 1 + j, 0, 1 − j} · IDFT → {1, 0, 0, 1} · Parseval 2 = 2 · saving 12 mult., 4 add.
P13	X(k) = {3, −j, 1, j} · IDFT → {1, 1, 1, 0} · Parseval 3 = 3 · saving 12 mult., 4 add.

SECTION 06

Sheet 2 Theory

Question 2.1

Questions

- ☐ 2.1 Prove that the DFT, X(k), of a sequence x(n) is periodic. p. 52

2.1

Prove that the DFT, X(k), of a sequence x(n) is periodic.

An important property of the DFT may be deduced if the kth component of the DFT, X(k), is compared with the $(k + N)^{th}$ component, X(k + N). Thus:

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\ e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N}$$

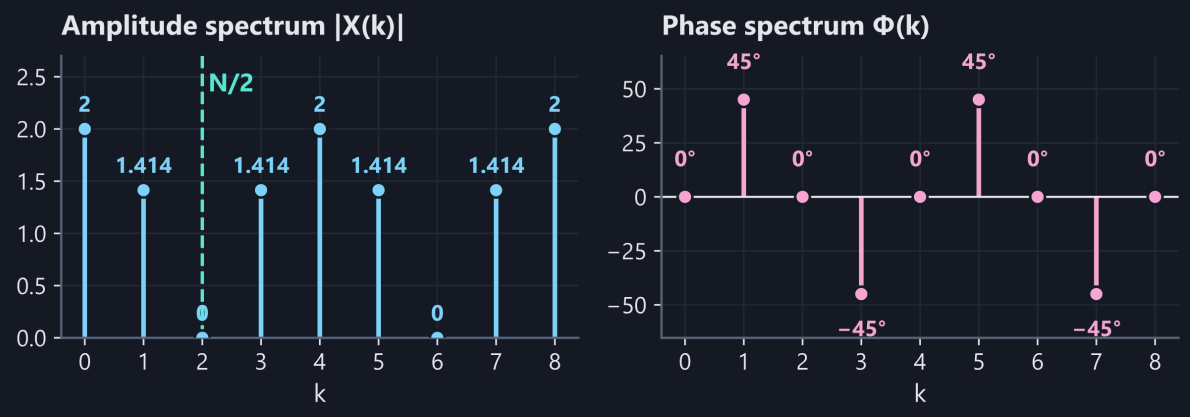
$$X(k + N) = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\ e^{-j2\pi kn/N} \cdot e^{-j2\pi n} = X(k)$$

since n is integral, so $e^{-j2\pi n} = 1$.

Sheet 2 Theory

The fact that **X(k + N) = X(k)** shows that the DFT is periodic with period N. This is the cyclical property of the DFT: the values of the DFT components are repetitive. If k = 0, then X(0) = X(N); e.g. for {1, 0, 0, 1}, X(0) = 2 and therefore X(4) = 2 also.

|X(k)| for {1, 0, 0, 1}: X(k + 4) = X(k), symmetric about N/2



Key idea لأن e^(−j2πn) = 1 لأي n صحيح، لما نعوّض k + N بنرجع لنفس القيمة، يعني الـ DFT periodic وطول الـ period بتاعه N.

Source — from your PDFs

Lecture	**LEC 5** · Sec. 2.1 · p. 4
Sheet	**Sheet 2** · Q 2.1

★ Cheat sheet · Sheet 2 Theory

Remember

2.1 e^{−j2πn} = 1, so X(k + N) = X(k): the DFT is periodic with period N.

SECTION 07

Sheet 2 Problems

Questions 2.2 to 2.6

Questions

- ☐ 2.2 DFT of {0, 1, 1, 0} and check by the IDFT p. 54
- ☐ 2.3 Verify Parseval's theorem for {0, 1, 1, 0} p. 57
- ☐ 2.4 FFT of {0, 1, 1, 0}, compare with the DFT, and IFFT p. 57
- ☐ 2.5 FFT of {0, 0, 1, 1, 1, 1, 0, 0}, spectra and IFFT p. 60
- ☐ 2.6 x(n) = cos(nπ/2): 8 samples, FFT and computational saving p. 63

2.2

DFT of {0, 1, 1, 0} and check by the IDFT

2.2 Calculate the DFT of the data sequence {0, 1, 1, 0} and check the validity of your answer by calculating its IDFT.

DFT (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \quad (2.4)$$

x(0) = 0, x(T) = 1, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 0 + 1 + 1 + 0 \\ &= 2 \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(1)n/4}$$
$$= 0 + e^{-j\pi/2} + e^{-j\pi} + 0$$
$$= [\cos(\pi/2) - j\,\sin(\pi/2)] + [\cos(\pi) - j\,\sin(\pi)]$$
$$= (-j) + (-1)$$
$$= \mathbf{-1 - j}$$

Thus X(1) = −1 − j, of magnitude √2 = 1.414 and phase angle Φ(1) = −135°.

With k = 2:

$$X(2) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(2)n/4}$$
$$= 0 + e^{-j\pi} + e^{-j2\pi} + 0$$
$$= [\cos(\pi) - j\,\sin(\pi)] + [\cos(2\pi) - j\,\sin(2\pi)]$$
$$= (-1) + 1$$
$$= \mathbf{0}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{3}\, x(nT)\, e^{-j2\pi(3)n/4}$$
$$= 0 + e^{-j3\pi/2} + e^{-j3\pi} + 0$$
$$= [\cos(3\pi/2) - j\,\sin(3\pi/2)] + [\cos(3\pi) - j\,\sin(3\pi)]$$
$$= j + (-1)$$
$$= \mathbf{-1 + j}$$

Thus X(3) = −1 + j, of magnitude √2 = 1.414 and phase angle Φ(3) = 135°.

So, the time series x(n) = {0, 1, 1, 0} has the DFT given by **X(k) = {2, −1 − j, 0, −1 + j}**.

IDFT check (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1}\, X(k)\, e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1}\, X(k)\, e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3}\, X(k)\, e^{j2\pi k(0)/4}$$
$$= \tfrac{1}{4}\,[2 + (-1 - j) + 0 + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

Sheet 2 Problems

With n = 1:

$$x(nT) = x(T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(1)/4}$$
$$= \tfrac{1}{4}\ [2 + (-1 - j)\ e^{j\pi/2} + 0 + (-1 + j)\ e^{j3\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j)(j) + (-1 + j)(-j)]$$
$$= \tfrac{1}{4}\ [2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(2)/4}$$
$$= \tfrac{1}{4}\ [2 + (-1 - j)\ e^{j\pi} + 0 + (-1 + j)\ e^{j3\pi}]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j)(-1) + (-1 + j)(-1)]$$
$$= \tfrac{1}{4}\ [2 + (1 + j) + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\ \Sigma_{k=0}^{3}\ X(k)\ e^{j2\pi k(3)/4}$$
$$= \tfrac{1}{4}\ [2 + (-1 - j)\ e^{j3\pi/2} + 0 + (-1 + j)\ e^{j9\pi/2}]$$
$$= \tfrac{1}{4}\ [2 + (-1 - j)(-j) + (-1 + j)(j)]$$
$$= \tfrac{1}{4}\ [2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{0, 1, 1, 0}**, the original sequence.

Result $X(k) = \{2, -1 - j, 0, -1 + j\}$ · IDFT → $x(n) = \{0, 1, 1, 0\}$

Key idea DFT بالتعويض المباشر زي مثال 2.1، وبعدين نرجع بالـ IDFT زي مثال 2.2 عشان نتأكد إن الـ sequence رجعت زي ما هي.

Key words DFT (as Example 2.1)

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Sheet	**Sheet 2** · Q 2.2

2.3

Verify Parseval's theorem for {0, 1, 1, 0}

2.3 Use the results of prob. (2.2) to verify Parseval's theorem for the data {0, 1, 1, 0}.

The normalized energy in the signal is given by either of the expressions:

$$\Sigma_{n=0}^{N-1} x^2(n) = (1/N) \Sigma_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$

$$L.H.S. = 0^2 + 1^2 + 1^2 + 0^2 = 2$$
$$R.H.S. = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = 2$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

Result $\Sigma x^2 = 2 = (1/4)(4 + 2 + 0 + 2) \rightarrow$ Parseval's theorem verified

Key idea نحسب مجموع x^2 في الوقت، ونقارنه بـ $(1/N)$ في مجموع $|X(k)|^2$ في التردد، والاتنين لازم يطلعوا قد بعض.

Source — from your PDFs

Lecture	**LEC 5** · Sec. 2.3 · p. 27
Sheet	**Sheet 2** · Q 2.3

2.4

FFT of {0, 1, 1, 0}, compare with the DFT, and IFFT

2.4 Calculate the DFT of the data sequence {0, 1, 1, 0} using the decimation-in-time FFT algorithm. Check the answer with that of Prob. (2.2). Compare numbers of complex additions and multiplications in the two methods. Calculate the IFFT to verify your answer.

FFT (as Example 2.3):

Sheet 2 Problems

This is a four-point DFT with x0 = 0, x1 = 1, x2 = 1, x3 = 0, and

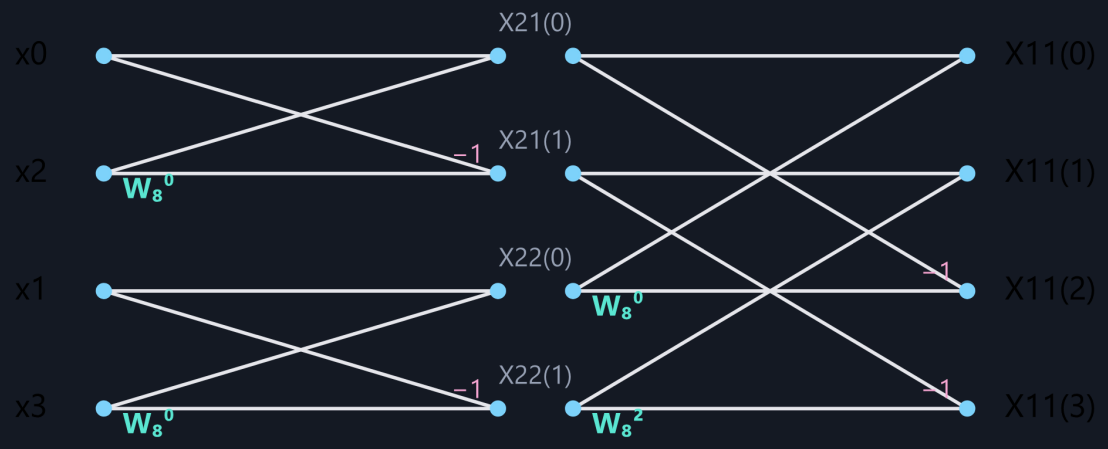
$$X1(k) = X11(k) + W_N^k\, X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 0 + 1 = 1$$
$$X21(1) = x0 - x2 = 0 - 1 = -1$$
$$X22(0) = x1 + x3 = 1 + 0 = 1$$
$$X22(1) = x1 - x3 = 1 - 0 = 1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = -1 + e^{-j\pi/2}(1) = -1 - j(1) = \mathbf{-1 - j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = -1 - e^{-j\pi/2}(1) = -1 + j(1) = \mathbf{-1 + j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values X(k) = {2, −1 − j, 0, −1 + j} are the same as those obtained in Prob. (2.2), but were more readily obtained using the FFT algorithm.

Comparison (Sec. 2.4, 2.5.2), N = 4:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

Sheet 2 Problems

IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \qquad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = -2 \qquad x22(1) = X(1) - X(3) = -j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\ x22(1)] = \tfrac{1}{4}[2 + j(-j2)] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\ x22(1)] = \tfrac{1}{4}[2 - j(-j2)] = \mathbf{0}$$

∴ x(n) = **{0, 1, 1, 0}**, the original data sequence.

Result FFT gives the same X(k) = {2, −1 − j, 0, −1 + j} · IFFT → x(n) = {0, 1, 1, 0}

Key idea نفس الـ sequence بالـ FFT، والمفروض يطلع نفس ناتج الـ DFT. وبعدين نرجع بالـ IFFT بالـ W السالبة ونقسم على 4.

Key words FFT (as Example 2.3) IFFT (Sec. 2.6)

Source — from your PDFs

Lecture	**LEC 6** · Example 2.3 · p. 16
Lecture	**LEC 6** · Sec. 2.5.2 · p. 18
Lecture	**LEC 6** · Sec. 2.6 · p. 20
Sheet	**Sheet 2** · Q 2.4

2.5

FFT of {0, 0, 1, 1, 1, 1, 0, 0}, spectra and IFFT

2.5 Calculate the FFT of the data sequence {0, 0, 1, 1, 1, 1, 0, 0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = x0 + x4 = 1$ $X21(1) = x0 - x4 = -1$	$X11(0) = X21(0) + W_8^0 X22(0) = 2$ $X11(1) = X21(1) + W_8^2 X22(1) = -1 - j$
$X22(0) = x2 + x6 = 1$ $X22(1) = x2 - x6 = 1$	$X11(2) = X21(0) - W_8^0 X22(0) = 0$ $X11(3) = X21(1) - W_8^2 X22(1) = -1 + j$
$X23(0) = x1 + x5 = 1$ $X23(1) = x1 - x5 = -1$	$X12(0) = X23(0) + W_8^0 X24(0) = 2$ $X12(1) = X23(1) + W_8^2 X24(1) = -1 - j$
$X24(0) = x3 + x7 = 1$ $X24(1) = x3 - x7 = 1$	$X12(2) = X23(0) - W_8^0 X24(0) = 0$ $X12(3) = X23(1) - W_8^2 X24(1) = -1 + j$

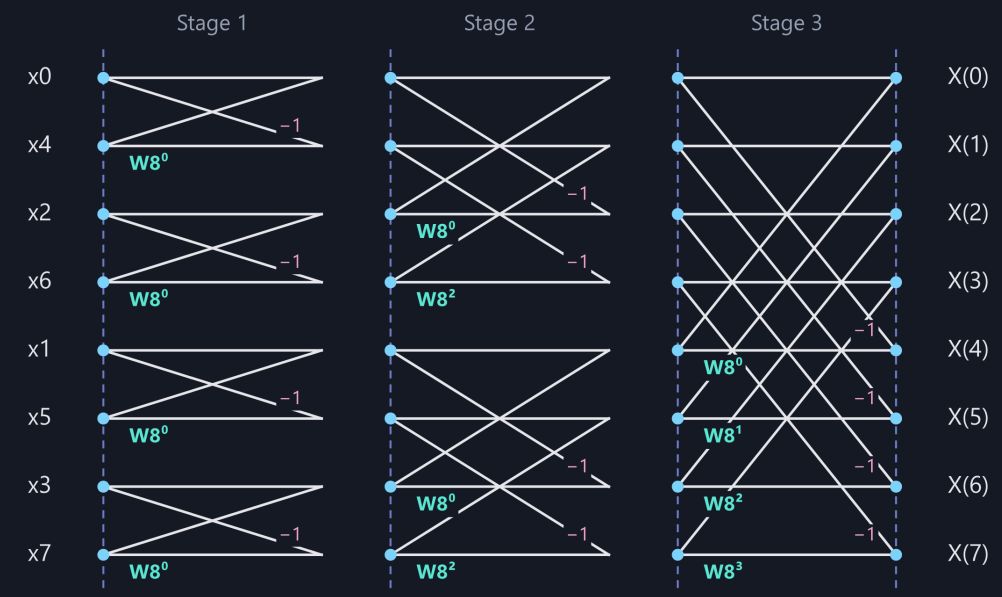
Stage 3: $X1(k) = X11(k) + W_8^k\, X12(k), \quad X1(k+4) = X11(k) - W_8^k\, X12(k), \quad k = 0, 1, 2, 3$

$$X(0) = 4 \qquad X(4) = 0$$
$$X(1) = -2.414 - j \qquad X(5) = 0.414 - j$$
$$X(2) = 0 \qquad X(6) = 0$$
$$X(3) = 0.414 + j \qquad X(7) = -2.414 + j$$

$\therefore X(k) =$ **{4, −2.414 − j, 0, 0.414 + j, 0, 0.414 − j, 0, −2.414 + j}**

Sheet 2 Problems

8-point DIT FFT: N/2 butterflies × $\log_2 N$ stages (Fig. 2.3)

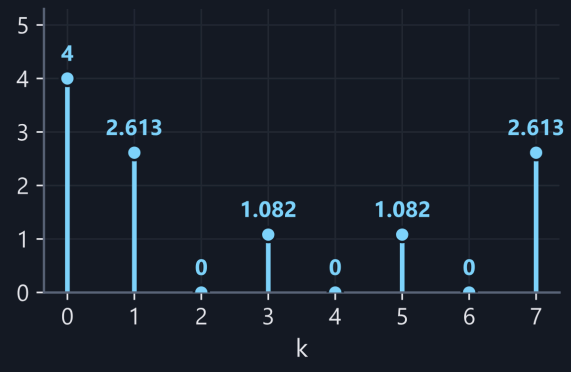


Amplitude and phase spectra ($|X(k)|$ and $\Phi(k)$ versus k; symmetrical about N/2 = 4):

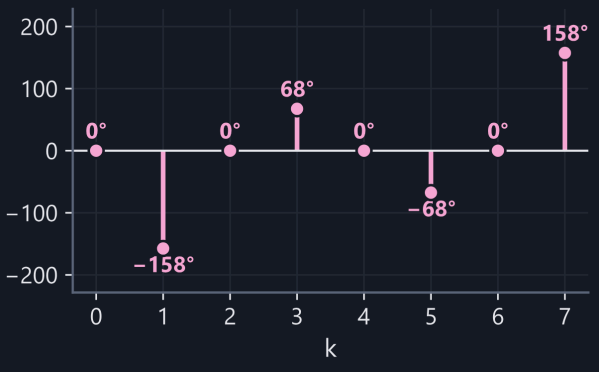
k	0	1	2	3	4	5	6	7
X(k)	4	−2.414 − j	0	0.414 + j	0	0.414 − j	0	−2.414 + j
$\|X(k)\|$	4	2.613	0	1.082	0	1.082	0	2.613
$\Phi(k)$	0°	−157.5°	0°	67.5°	0°	−67.5°	0°	157.5°

Spectra of {0, 0, 1, 1, 1, 1, 0, 0}

Amplitude spectrum |X(k)|



Phase spectrum Φ(k)



IFFT check (same flowgraph, W_8^{-k}, divided by N = 8):

Sheet 2 Problems

Re-ordered input (Table 2.1): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7). $W_8^{0} = 1$, $W_8^{-1} = (1 + j)/\sqrt{2}$, $W_8^{-2} = +j$, $W_8^{-3} = (-1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = X(0) + X(4) = 4$ $X21(1) = X(0) - X(4) = 4$	$X11(0) = X21(0) + W_8^{0}X22(0) = 4$ $X11(1) = X21(1) + W_8^{-2}X22(1) = 4$
$X22(0) = X(2) + X(6) = 0$ $X22(1) = X(2) - X(6) = 0$	$X11(2) = X21(0) - W_8^{0}X22(0) = 4$ $X11(3) = X21(1) - W_8^{-2}X22(1) = 4$
$X23(0) = X(1) + X(5) = -2 - j2$ $X23(1) = X(1) - X(5) = -2.828$	$X12(0) = X23(0) + W_8^{0}X24(0) = -4$ $X12(1) = X23(1) + W_8^{-2}X24(1) = -2.828 + j2.828$
$X24(0) = X(3) + X(7) = -2 + j2$ $X24(1) = X(3) - X(7) = 2.828$	$X12(2) = X23(0) - W_8^{0}X24(0) = -j4$ $X12(3) = X23(1) - W_8^{-2}X24(1) = -2.828 - j2.828$

Stage 3: $X1(k) = X11(k) + W_8^{-k}\, X12(k), \quad X1(k+4) = X11(k) - W_8^{-k}\, X12(k), \quad k = 0, 1, 2, 3$

$$x(0) = (1/8)(0) = \mathbf{0} \qquad x(4) = (1/8)(8) = \mathbf{1}$$
$$x(1) = (1/8)(0) = \mathbf{0} \qquad x(5) = (1/8)(8) = \mathbf{1}$$
$$x(2) = (1/8)(8) = \mathbf{1} \qquad x(6) = (1/8)(0) = \mathbf{0}$$
$$x(3) = (1/8)(8) = \mathbf{1} \qquad x(7) = (1/8)(0) = \mathbf{0}$$

∴ x(n) = **{0, 0, 1, 1, 1, 1, 0, 0}**, the original data sequence.

Result $X(k) = \{4, -2.414 - j, 0, 0.414 + j, 0, 0.414 - j, 0, -2.414 + j\}$ · $|X(k)| = \{4, 2.613, 0, 1.082, 0, 1.082, 0, 2.613\}$ · IFFT → {0, 0, 1, 1, 1, 1, 0, 0}

Key idea FFT لـ 8 نقط: نرتب الـ input بالـ bit reversal (x0, x4, x2, x6, x1, x5, x3, x7)، ونعمل 3 stages من الـ butterflies بقيم W_8.

Key words FFT (8-point, Fig. 2.3)

Sheet 2 Problems

Source — from your PDFs

Lecture	**LEC 6** · Table 2.1 · p. 10
Lecture	**LEC 6** · Sec. 2.6 · p. 20
Sheet	**Sheet 2** · Q 2.5

2.6

x(n) = cos(nπ/2): 8 samples, FFT and computational saving

2.6 For the signal x(n) = cos(nπ/2), find the first 8 samples and then, using the FFT algorithm, calculate the frequency components X(k). Assume we use the DFT algorithm instead of the FFT algorithm, what is the computational saving?

$$x(n) = \cos(n\pi/2): \quad x(0) = \cos 0 = 1,\ x(1) = \cos(\pi/2) = 0,\ x(2) = \cos \pi = -1,\ x(3) = \cos(3\pi/2) = 0,\ \ldots$$
$$x(n) = \{1, 0, -1, 0, 1, 0, -1, 0\}$$

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
X21(0) = x0 + x4 = 2 X21(1) = x0 − x4 = 0	X11(0) = X21(0) + W_8^0X22(0) = 0 X11(1) = X21(1) + W_8^2X22(1) = 0
X22(0) = x2 + x6 = −2 X22(1) = x2 − x6 = 0	X11(2) = X21(0) − W_8^0X22(0) = 4 X11(3) = X21(1) − W_8^2X22(1) = 0
X23(0) = x1 + x5 = 0 X23(1) = x1 − x5 = 0	X12(0) = X23(0) + W_8^0X24(0) = 0 X12(1) = X23(1) + W_8^2X24(1) = 0
X24(0) = x3 + x7 = 0 X24(1) = x3 − x7 = 0	X12(2) = X23(0) − W_8^0X24(0) = 0 X12(3) = X23(1) − W_8^2X24(1) = 0

Stage 3: $X1(k) = X11(k) + W_8^k\, X12(k), \quad X1(k+4) = X11(k) - W_8^k\, X12(k), \quad k = 0, 1, 2, 3$

Sheet 2 Problems

```
X(0) = 0      X(4) = 0
X(1) = 0      X(5) = 0
X(2) = 4      X(6) = 4
X(3) = 0      X(7) = 0
```

∴ X(k) = **{0, 0, 4, 0, 0, 0, 4, 0}**

k	0	1	2	3	4	5	6	7
X(k)	0	0	4	0	0	0	4	0
|X(k)|	0	0	4	0	0	0	4	0
Φ(k)	0°	0°	0°	0°	0°	0°	0°	0°

Spectra of x(n) = cos(nπ/2)



Computational saving (Sec. 2.5.2), N = 8:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 64$	$(N/2)\log_2 N = 12$	**52**
No. of complex additions	$N(N-1) = 56$	$N\log_2 N = 24$	**32**

Result x(n) = {1, 0, −1, 0, 1, 0, −1, 0} · X(k) = {0, 0, 4, 0, 0, 0, 4, 0} · saving 52 mult., 32 add.

Key idea نطلع 8 samples من cos(nπ/2)، ونعمل FFT لـ 8 نقط، فبيطلع تردد واحد عند k = 2 وصورته عند k = 6. والـ saving بيبقى 52 ضربة.

Key words FFT (8-point, Fig. 2.3)

Source — from your PDFs

Lecture	**LEC 6** · Sec. 2.5 · p. 6
Lecture	**LEC 6** · Sec. 2.5.2 · p. 18
Sheet	**Sheet 2** · Q 2.6

★ Cheat sheet · Sheet 2 Problems

Final results

2.2	X(k) = {2, −1 − j, 0, −1 + j} · IDFT → x(n) = {0, 1, 1, 0}
2.3	$\Sigma x^2 = 2 = (1/4)(4 + 2 + 0 + 2)$ → Parseval's theorem verified
2.4	FFT gives the same X(k) = {2, −1 − j, 0, −1 + j} · IFFT → x(n) = {0, 1, 1, 0}
2.5	X(k) = {4, −2.414 − j, 0, 0.414 + j, 0, 0.414 − j, 0, −2.414 + j} · |X(k)| = {4, 2.613, 0, 1.082, 0, 1.082, 0, 2.613} · IFFT → {0, 0, 1, 1, 1, 1, 0, 0}
2.6	x(n) = {1, 0, −1, 0, 1, 0, −1, 0} · X(k) = {0, 0, 4, 0, 0, 0, 4, 0} · saving 52 mult., 32 add.

SECTION 08

Assignment

DSP Lecture Assignment, Fall 27

Questions

☐ Q DSP Lecture Assignment — Fall 27 p. 66

Q

DSP Lecture Assignment — Fall 27

*For the sequence **x(n) = {0, 2, 1, 1, 3, 1, 2, 0}**, calculate the frequency components X(k) by the DFT method and check the results by calculating x(n) by the IDFT method. Repeat the calculation of X(k) by the FFT algorithm and check by calculating x(n) by the IFFT algorithm.*

(1) Frequency components X(k) by the DFT method

The DFT of an N-point sequence is given by Eq. (2.4):

$$X(k) = F_D[x(nT)] = \sum_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT}, \quad k = 0, 1, \ldots, N-1 \qquad (2.4)$$
$$\text{with } \Omega = 2\pi/NT, \text{ so that } k\Omega nT = 2\pi kn/N$$

Here N = 8 and x(0) = 0, x(T) = 2, x(2T) = 1, x(3T) = 1, x(4T) = 3, x(5T) = 1, x(6T) = 2, x(7T) = 0. It is required to find X(k) for k = 0, 1, ..., 7. The exponent is $-j2\pi kn/8 = -j\pi kn/4$; the term n = 0 vanishes because x(0) = 0. Values used: $e^{\mp j\pi/4} = 0.707 \mp j0.707$, $e^{\mp j\pi/2} = \mp j$, $e^{\mp j3\pi/4} = -0.707 \mp j0.707$, $e^{\mp j\pi} = -1$.

With k = 0:

$$X(0) = \sum_{n=0}^{7} x(nT)\, e^{-j2\pi(0)n/8} = \sum_{n=0}^{7} x(nT)\, e^{-j0\cdot n}$$
$$= 0 + 2 + 1 + 1 + 3 + 1 + 2$$
$$= 2 + 1 + 1 + 3 + 1 + 2$$
$$= 2 + 1 + 1 + 3 + 1 + 2$$
$$= \mathbf{10}$$

Thus X(0) = 10, of magnitude 10 and phase angle Φ(0) = 0°.

With k = 1:

$$X(1) = \sum_{n=0}^{7} x(nT)\, e^{-j2\pi(1)n/8} = \sum_{n=0}^{7} x(nT)\, e^{-j\pi/4\cdot n}$$
$$= 0 + 2e^{-j\pi/4} + e^{-j\pi/2} + e^{-j3\pi/4} + 3e^{-j\pi} + e^{j3\pi/4} + 2e^{j\pi/2}$$
$$= 2(0.707 - j0.707) + (-j) + (-0.707 - j0.707) + 3(-1) + (-0.707 + j0.707) + 2(j)$$
$$= (1.414 - j1.414) + (-j) + (-0.707 - j0.707) + (-3) + (-0.707 + j0.707) + j2$$
$$= \mathbf{-3 - j0.414}$$

Thus X(1) = −3 − j0.414, of magnitude 3.028 and phase angle Φ(1) = −172.1°.

Assignment

With k = 2:

$$X(2) = \Sigma_{n=0}^{7} x(nT)\, e^{-j2\pi(2)n/8} = \Sigma_{n=0}^{7} x(nT)\, e^{-j\pi/2 \cdot n}$$
$$= 0 + 2e^{-j\pi/2} + e^{-j\pi} + e^{j\pi/2} + 3 + e^{-j\pi/2} + 2e^{-j\pi}$$
$$= 2(-j) + (-1) + (j) + 3 + (-j) + 2(-1)$$
$$= (-j2) + (-1) + j + 3 + (-j) + (-2)$$
$$= \mathbf{-j2}$$

Thus X(2) = −j2, of magnitude 2 and phase angle Φ(2) = −90°.

With k = 3:

$$X(3) = \Sigma_{n=0}^{7} x(nT)\, e^{-j2\pi(3)n/8} = \Sigma_{n=0}^{7} x(nT)\, e^{-j3\pi/4 \cdot n}$$
$$= 0 + 2e^{-j3\pi/4} + e^{j\pi/2} + e^{-j\pi/4} + 3e^{-j\pi} + e^{j\pi/4} + 2e^{-j\pi/2}$$
$$= 2(-0.707 - j0.707) + (j) + (0.707 - j0.707) + 3(-1) + (0.707 + j0.707) + 2(-j)$$
$$= (-1.414 - j1.414) + j + (0.707 - j0.707) + (-3) + (0.707 + j0.707) + (-j2)$$
$$= \mathbf{-3 - j2.414}$$

Thus X(3) = −3 − j2.414, of magnitude 3.851 and phase angle Φ(3) = −141.2°.

With k = 4:

$$X(4) = \Sigma_{n=0}^{7} x(nT)\, e^{-j2\pi(4)n/8} = \Sigma_{n=0}^{7} x(nT)\, e^{-j\pi \cdot n}$$
$$= 0 + 2e^{-j\pi} + 1 + e^{-j\pi} + 3 + e^{-j\pi} + 2$$
$$= 2(-1) + 1 + (-1) + 3 + (-1) + 2$$
$$= (-2) + 1 + (-1) + 3 + (-1) + 2$$
$$= \mathbf{2}$$

Thus X(4) = 2, of magnitude 2 and phase angle Φ(4) = 0°.

With k = 5:

$$X(5) = \Sigma_{n=0}^{7} x(nT)\, e^{-j2\pi(5)n/8} = \Sigma_{n=0}^{7} x(nT)\, e^{-j-3\pi/4 \cdot n}$$
$$= 0 + 2e^{j3\pi/4} + e^{-j\pi/2} + e^{j\pi/4} + 3e^{-j\pi} + e^{-j\pi/4} + 2e^{j\pi/2}$$
$$= 2(-0.707 + j0.707) + (-j) + (0.707 + j0.707) + 3(-1) + (0.707 - j0.707) + 2(j)$$
$$= (-1.414 + j1.414) + (-j) + (0.707 + j0.707) + (-3) + (0.707 - j0.707) + j2$$
$$= \mathbf{-3 + j2.414}$$

Thus X(5) = −3 + j2.414, of magnitude 3.851 and phase angle Φ(5) = 141.2°.

With k = 6:

$$X(6) = \Sigma_{n=0}^{7} x(nT)\, e^{-j2\pi(6)n/8} = \Sigma_{n=0}^{7} x(nT)\, e^{-j-\pi/2\cdot n}$$
$$= 0 + 2e^{j\pi/2} + e^{-j\pi} + e^{-j\pi/2} + 3 + e^{j\pi/2} + 2e^{-j\pi}$$
$$= 2(j) + (-1) + (-j) + 3 + (j) + 2(-1)$$
$$= j2 + (-1) + (-j) + 3 + j + (-2)$$
$$= \mathbf{j2}$$

Thus X(6) = j2, of magnitude 2 and phase angle Φ(6) = 90°.

With k = 7:

$$X(7) = \Sigma_{n=0}^{7} x(nT)\, e^{-j2\pi(7)n/8} = \Sigma_{n=0}^{7} x(nT)\, e^{-j-\pi/4\cdot n}$$
$$= 0 + 2e^{j\pi/4} + e^{j\pi/2} + e^{j3\pi/4} + 3e^{-j\pi} + e^{-j3\pi/4} + 2e^{-j\pi/2}$$
$$= 2(0.707 + j0.707) + (j) + (-0.707 + j0.707) + 3(-1) + (-0.707 - j0.707) + 2(-j)$$
$$= (1.414 + j1.414) + j + (-0.707 + j0.707) + (-3) + (-0.707 - j0.707) + (-j2)$$
$$= \mathbf{-3 + j0.414}$$

Thus X(7) = −3 + j0.414, of magnitude 3.028 and phase angle Φ(7) = 172.1°.

So, the time series x(n) = {0, 2, 1, 1, 3, 1, 2, 0} has the DFT

$$\mathbf{X(k) = \{10, -3 - j0.414, -j2, -3 - j2.414, 2, -3 + j2.414, j2, -3 + j0.414\}}$$

k	0	1	2	3	4	5	6	7
X(k)	10	−3 − j0.414	−j2	−3 − j2.414	2	−3 + j2.414	j2	−3 + j0.414
|X(k)|	10	3.028	2	3.851	2	3.851	2	3.028
Φ(k)	0°	−172.1°	−90°	−141.2°	0°	141.2°	90°	172.1°

Note: X(8 − k) = X*(k) (e.g. X(7) = X*(1)), as expected for a real sequence.

(2) Check: original sequence x(n) by the IDFT method

The IDFT is given by Eq. (2.7):

$$x(nT) = F_D^{-1}[X(k)] = (1/N) \Sigma_{k=0}^{N-1} X(k)\, e^{jk\Omega nT} = (1/8) \Sigma_{k=0}^{7} X(k)\, e^{j\pi kn/4} \qquad (2.7)$$

With n = 0:

$$x(0T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(0)/8}$$
$$= (1/8)[10 + (-3 - j0.414) + (-j2) + (-3 - j2.414) + 2 + (-3 + j2.414) + j2 + (-3 + j0.414)]$$
$$= (1/8)(0) = \mathbf{0}$$

With n = 1:

$$x(1T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(1)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j\pi/4} + (-j2)e^{j\pi/2} + (-3 - j2.414)e^{j3\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{-j3\pi/4} + j2e^{-j\pi/2} + (-3 + j0.414)e^{-j\pi/4}]$$
$$= (1/8)[10 + (-1.828 - j2.414) + 2 + (3.828 - j0.414) + (-2) + (3.828 + j0.414) + 2 + (-1.828 + j2.414)]$$
$$= (1/8)(16) = \mathbf{2}$$

With n = 2:

$$x(2T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(2)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j\pi/2} + (-j2)e^{j\pi} + (-3 - j2.414)e^{-j\pi/2} + 2 + (-3 + j2.414)e^{j\pi/2} + j2e^{j\pi} + (-3 + j0.414)e^{-j\pi/2}]$$
$$= (1/8)[10 + (0.414 - j3) + j2 + (-2.414 + j3) + 2 + (-2.414 - j3) + (-j2) + (0.414 + j3)]$$
$$= (1/8)(8) = \mathbf{1}$$

With n = 3:

$$x(3T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(3)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j3\pi/4} + (-j2)e^{-j\pi/2} + (-3 - j2.414)e^{j\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{-j\pi/4} + j2e^{j\pi/2} + (-3 + j0.414)e^{-j3\pi/4}]$$
$$= (1/8)[10 + (2.414 - j1.828) + (-2) + (-0.414 - j3.828) + (-2) + (-0.414 + j3.828) + (-2) + (2.414 + j1.828)]$$
$$= (1/8)(8) = \mathbf{1}$$

With n = 4:

$$x(4T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(4)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{j\pi} + (-j2) + (-3 - j2.414)e^{j\pi} + 2 + (-3 + j2.414)e^{j\pi} + j2 + (-3 + j0.414)e^{j\pi}]$$
$$= (1/8)[10 + (3 + j0.414) + (-j2) + (3 + j2.414) + 2 + (3 - j2.414) + j2 + (3 - j0.414)]$$
$$= (1/8)(24) = \mathbf{3}$$

With n = 5:

$$x(5T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(5)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{-j3\pi/4} + (-j2)e^{j\pi/2} + (-3 - j2.414)e^{-j\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{j\pi/4} + j2e^{-j\pi/2} + (-3 + j0.414)e^{j3\pi/4}]$$
$$= (1/8)[10 + (1.828 + j2.414) + 2 + (-3.828 + j0.414) + (-2) + (-3.828 - j0.414) + 2 + (1.828 - j2.414)]$$
$$= (1/8)(8) = \mathbf{1}$$

With n = 6:

$$x(6T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(6)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{-j\pi/2} + (-j2)e^{j\pi} + (-3 - j2.414)e^{j\pi/2} + 2 + (-3 + j2.414)e^{-j\pi/2} + j2e^{j\pi} + (-3 + j0.414)e^{j\pi/2}]$$
$$= (1/8)[10 + (-0.414 + j3) + j2 + (2.414 - j3) + 2 + (2.414 + j3) + (-j2) + (-0.414 - j3)]$$
$$= (1/8)(16) = \mathbf{2}$$

With n = 7:

$$x(7T) = (1/8)\ \Sigma_{k=0}^{7}\ X(k)\ e^{j2\pi k(7)/8}$$
$$= (1/8)[10 + (-3 - j0.414)e^{-j\pi/4} + (-j2)e^{-j\pi/2} + (-3 - j2.414)e^{-j3\pi/4} + 2e^{j\pi} + (-3 + j2.414)e^{j3\pi/4} + j2e^{j\pi/2} + (-3 + j0.414)e^{j\pi/4}]$$
$$= (1/8)[10 + (-2.414 + j1.828) + (-2) + (0.414 + j3.828) + (-2) + (0.414 - j3.828) + (-2) + (-2.414 - j1.828)]$$
$$= (1/8)(0) = \mathbf{0}$$

∴ x(n) = **{0, 2, 1, 1, 3, 1, 2, 0}**, which is the original data sequence, so the DFT results are correct.

(3) Repeat: X(k) by the FFT algorithm (decimation in time)

For N = 8 = 2^3 the DFT is decomposed into $\log_2 8 = 3$ stages: the 8-point DFT X1(k) is expressed by Eq. (2.24) in terms of the 4-point DFTs X11(k) (even samples) and X12(k) (odd samples), and each of these in terms of 2-point DFTs X21(k) ... X24(k), as in Table 2.1.

$$X1(k) = X11(k) + W_N^{k}\ X12(k), \quad W_N = e^{-j2\pi/N} \qquad (2.24)$$

Re-ordering (Table 2.1, line 6): x0, x4, x2, x6, x1, x5, x3, x7, where x0 = 0, x1 = 2, x2 = 1, x3 = 1, x4 = 3, x5 = 1, x6 = 2, x7 = 0.

Weights: $W_8^0 = 1$, $W_8^1 = e^{-j\pi/4} = 0.707 - j0.707$, $W_8^2 = -j$, $W_8^3 = e^{-j3\pi/4} = -0.707 - j0.707$.

Stage 1 — four 2-point DFTs (the weight is $W_8^0 = 1$):

$$X21(0) = x0 + x4 = 3, \quad X21(1) = x0 - x4 = -3$$
$$X22(0) = x2 + x6 = 3, \quad X22(1) = x2 - x6 = -1$$
$$X23(0) = x1 + x5 = 3, \quad X23(1) = x1 - x5 = 1$$
$$X24(0) = x3 + x7 = 1, \quad X24(1) = x3 - x7 = 1$$

Stage 2 — two 4-point DFTs, Eqs. (2.25a–d), with $W_8^2 = e^{-j\pi/2} = -j$:

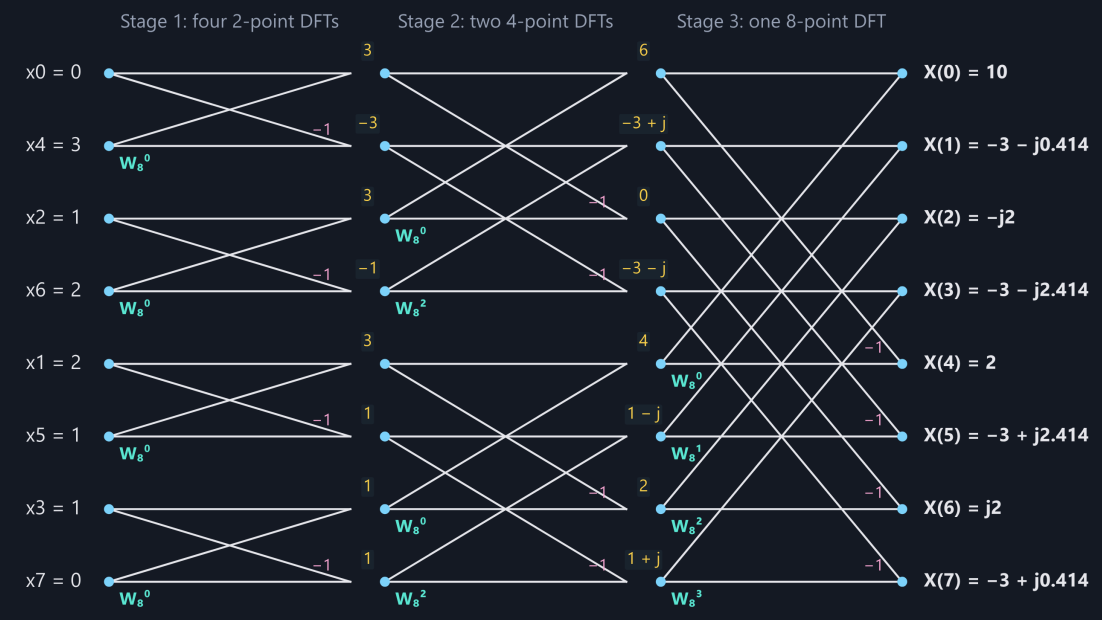
$$X11(0) = X21(0) + W_8^0\, X22(0) = 3 + 3 = 6$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = (-3) + (-j)(-1) = -3 + j$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 3 - 3 = 0$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = (-3) - (-j)(-1) = -3 - j$$

$$X12(0) = X23(0) + W_8^0\, X24(0) = 3 + 1 = 4$$
$$X12(1) = X23(1) + W_8^2\, X24(1) = 1 + (-j)1 = 1 - j$$
$$X12(2) = X23(0) - W_8^0\, X24(0) = 3 - 1 = 2$$
$$X12(3) = X23(1) - W_8^2\, X24(1) = 1 - (-j)1 = 1 + j$$

Stage 3 — the 8-point DFT, Eq. (2.24): $X1(k) = X11(k) + W_8^k\, X12(k)$, and $X1(k + 4) = X11(k) - W_8^k\, X12(k)$, $k = 0, 1, 2, 3$:

$$X1(0) = X11(0) + W_8^0\, X12(0) = 6 + 14 = 6 + 4 = \mathbf{10}$$
$$X1(1) = X11(1) + W_8^1\, X12(1) = (-3 + j) + (0.707 - j0.707)(1 - j) = (-3 + j) + (-j1.414) = \mathbf{-3 - j0.414}$$
$$X1(2) = X11(2) + W_8^2\, X12(2) = 0 + (-j)2 = 0 + (-j2) = \mathbf{-j2}$$
$$X1(3) = X11(3) + W_8^3\, X12(3) = (-3 - j) + (-0.707 - j0.707)(1 + j) = (-3 - j) + (-j1.414) = \mathbf{-3 - j2.414}$$
$$X1(4) = X11(0) - W_8^0\, X12(0) = 6 - 4 = \mathbf{2}$$
$$X1(5) = X11(1) - W_8^1\, X12(1) = (-3 + j) - (-j1.414) = \mathbf{-3 + j2.414}$$
$$X1(6) = X11(2) - W_8^2\, X12(2) = 0 - (-j2) = \mathbf{j2}$$
$$X1(7) = X11(3) - W_8^3\, X12(3) = (-3 - j) - (-j1.414) = \mathbf{-3 + j0.414}$$

Assignment



∴ X(k) = **{10, −3 − j0.414, −j2, −3 − j2.414, 2, −3 + j2.414, j2, −3 + j0.414}** — the same values obtained by the DFT method.

(4) Check: x(n) by the IFFT algorithm

Comparing Eq. (2.27) with Eq. (2.26), the IFFT differs from the FFT only in the factor 1/N and the sign of the exponent. So the same flowgraph is used with input X(k) (re-ordered), weights W_8^{-k} instead of W_8^{k}, and the outputs are divided by N = 8:

$$x(\mu) = (1/N)\ \Sigma_{\lambda=0}^{N-1}\ X(\lambda)\ e^{+j\mu(2\pi/N)\lambda}, \quad \mu = 0,\ 1,\ \ldots,\ N-1 \qquad (2.27)$$

Re-ordering (Table 2.1, line 6): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7), where X(0) = 10, X(1) = −3 − j0.414, X(2) = −j2, X(3) = −3 − j2.414, X(4) = 2, X(5) = −3 + j2.414, X(6) = j2, X(7) = −3 + j0.414.

Weights: $W_8^{0} = 1$, $W_8^{-1} = e^{j\pi/4} = 0.707 + j0.707$, $W_8^{-2} = +j$, $W_8^{-3} = e^{j3\pi/4} = -0.707 + j0.707$.

Stage 1 — four 2-point DFTs (the weight is $W_8^{0} = 1$):

$$X21(0) = X(0) + X(4) = 12, \quad X21(1) = X(0) - X(4) = 8$$
$$X22(0) = X(2) + X(6) = 0, \quad X22(1) = X(2) - X(6) = -j4$$
$$X23(0) = X(1) + X(5) = -6 + j2, \quad X23(1) = X(1) - X(5) = -j2.828$$
$$X24(0) = X(3) + X(7) = -6 - j2, \quad X24(1) = X(3) - X(7) = -j2.828$$

Stage 2 — two 4-point DFTs, Eqs. (2.25a–d), with $W_8^{-2} = e^{+j\pi/2} = +j$:

$$X11(0) = X21(0) + W_8^{0}\, X22(0) = 12 + 0 = 12$$
$$X11(1) = X21(1) + W_8^{-2}\, X22(1) = 8 + (j)(-j4) = 12$$
$$X11(2) = X21(0) - W_8^{0}\, X22(0) = 12 - 0 = 12$$
$$X11(3) = X21(1) - W_8^{-2}\, X22(1) = 8 - (j)(-j4) = 4$$

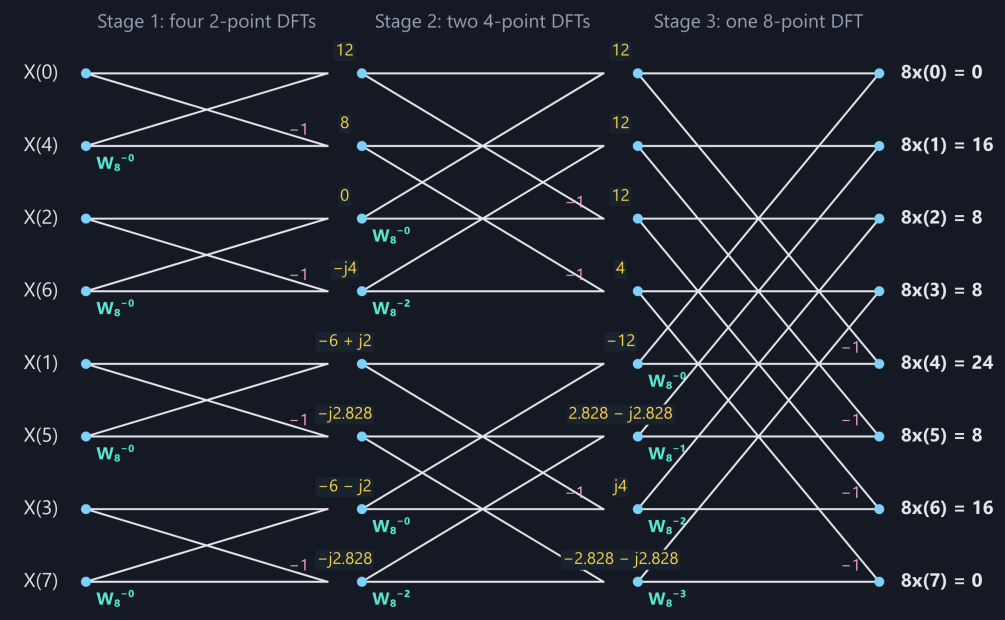
$$X12(0) = X23(0) + W_8^{0}\, X24(0) = (-6 + j2) + (-6 - j2) = -12$$
$$X12(1) = X23(1) + W_8^{-2}\, X24(1) = (-j2.828) + (j)(-j2.828) = 2.828 - j2.828$$
$$X12(2) = X23(0) - W_8^{0}\, X24(0) = (-6 + j2) - (-6 - j2) = j4$$
$$X12(3) = X23(1) - W_8^{-2}\, X24(1) = (-j2.828) - (j)(-j2.828) = -2.828 - j2.828$$

Stage 3 — the 8-point DFT, Eq. (2.24): $X1(k) = X11(k) + W_8^{-k}\, X12(k)$, and $X1(k + 4) = X11(k) - W_8^{-k}\, X12(k)$, $k = 0, 1, 2, 3$:

$$X1(0) = X11(0) + W_8^{-0}\, X12(0) = 12 + 1(-12) = 12 + (-12) = \mathbf{0}$$
$$X1(1) = X11(1) + W_8^{-1}\, X12(1) = 12 + (0.707 + j0.707)(2.828 - j2.828) = 12 + 4 = \mathbf{16}$$
$$X1(2) = X11(2) + W_8^{-2}\, X12(2) = 12 + jj4 = 12 + (-4) = \mathbf{8}$$
$$X1(3) = X11(3) + W_8^{-3}\, X12(3) = 4 + (-0.707 + j0.707)(-2.828 - j2.828) = 4 + 4 = \mathbf{8}$$
$$X1(4) = X11(0) - W_8^{-0}\, X12(0) = 12 - (-12) = \mathbf{24}$$
$$X1(5) = X11(1) - W_8^{-1}\, X12(1) = 12 - 4 = \mathbf{8}$$
$$X1(6) = X11(2) - W_8^{-2}\, X12(2) = 12 - (-4) = \mathbf{16}$$
$$X1(7) = X11(3) - W_8^{-3}\, X12(3) = 4 - 4 = \mathbf{0}$$

Assignment

IFFT flowgraph: same butterflies, W_8^{-k}, outputs divided by N = 8



$$x(0) = (1/8)(0) = 0$$
$$x(1) = (1/8)(16) = 2$$
$$x(2) = (1/8)(8) = 1$$
$$x(3) = (1/8)(8) = 1$$
$$x(4) = (1/8)(24) = 3$$
$$x(5) = (1/8)(8) = 1$$
$$x(6) = (1/8)(16) = 2$$
$$x(7) = (1/8)(0) = 0$$

∴ x(n) = **{0, 2, 1, 1, 3, 1, 2, 0}**, the original data sequence, so the FFT results are correct.

Final answer

X(k) = {10, −3 − j0.414, −j2, −3 − j2.414, 2, −3 + j2.414, j2, −3 + j0.414}

DFT and FFT give the same X(k); IDFT and IFFT both return x(n) = {0, 2, 1, 1, 3, 1, 2, 0}

Result X(k) = {10, −3 − j0.414, −j2, −3 − j2.414, 2, −3 + j2.414, j2, −3 + j0.414} · IDFT / IFFT → {0, 2, 1, 1, 3, 1, 2, 0}

Assignment

Key idea مسألة 8 نقط كاملة: DFT أو FFT، وبعدين IDFT أو IFFT عشان نرجع للـ sequence. الـ W_8 بتاخد 4 قيم بس: 1، و$j0.707$ – 0.707، وز–، و–0.707 – $j0.707$.

Source — from your PDFs

Lecture	**LEC 5** · Example 2.1 · p. 11
Lecture	**LEC 5** · Example 2.2 · p. 23
Lecture	**LEC 6** · Table 2.1 · p. 10
Lecture	**LEC 6** · Sec. 2.6 · p. 20
Assignment	**DSP Lecture Assignment — Fall 27**